Repairing Privately Generated Synthetic Data with Integrity and Statistical Considerations

ABSTRACT

Differentially private (DP) synthetic data can be shared freely for downstream analysis, but an already generated dataset may still violate integrity constraints or distort statistics that are important for analysis. Regenerating the data is often costly, may require a substantial privacy-budget allocation, and does not preserve the released sample. Existing repair methods, meanwhile, focus on constraint satisfaction and ignore the statistical targets that motivated synthetic data generation.

We introduce the *synthetic data repair problem*: given a DP synthetic dataset, a set of Denial Constraints (DCs), and noisy marginal targets derived from the private data, delete tuples so that all DCs are satisfied while minimizing a weighted objective that trades off data retention against relative marginal error. This soft formulation keeps the problem feasible while capturing the requirements of reliable synthetic data: maintain the largest sample with logical consistency and fidelity to selected private data statistics. We show that the problem is NP-hard even in restricted settings. We then present the first framework for post-hoc repair of DP synthetic data under both integrity and statistical considerations. The framework takes marginal targets as input and formulates repair as a mixed-integer linear program that yields an optimal solution. Since exact optimization can be expensive at scale, we also develop a practical two-phase greedy heuristic. We also describe one possible differentially private procedure for acquiring such targets when they are not already available. Experiments on four real-world datasets and state-of-the-art DP synthesizers show that our approach eliminates DC violations while preserving statistical fidelity, offering a practical tradeoff among integrity, accuracy, and scalability.

1 INTRODUCTION

Differential privacy (DP) is widely regarded as the gold standard for protecting individual privacy, offering strong, quantifiable, and composable guarantees for data analysis and release [15, 17]. A particularly powerful application of DP is synthetic data generation. By virtue of the post-processing property of DP [17], DP synthetic datasets can be freely shared and analyzed without additional privacy loss, supporting tasks such as exploratory analysis, model training, and query answering over sensitive data. A large body of work has developed DP synthetic data generation frameworks based on diverse techniques, ranging from deep generative models [52, 54, 57] to statistical modeling approaches [24, 25, 40, 41, 58]. Despite their differences, these methods share a common goal of preserving the statistical characteristics of the private data, most notably through the use of noisy low-dimensional marginals, typically two-way marginals, to guide synthetic data generation [40, 42, 58].

In addition to statistical fidelity, synthetic data may be required to satisfy integrity constraints that encode domain knowledge and semantic validity. Denial Constraints (DCs) [7, 50] form the most general and widely adopted class of such constraints, subsuming functional dependencies and other common restrictions by forbidding disallowed patterns across tuples. Recent work has begun to address this need, most notably through Kamino [21], which generates DP synthetic data that satisfies a specified set of DCs.

However, existing systems such as Kamino focus exclusively on the *generation* of synthetic data from scratch. In many realistic settings, synthetic data may already have been generated, often at substantial computational cost, and only later discovered to be inconsistent with newly imposed or previously overlooked integrity constraints. At the same time, synthetic data is increasingly expected to satisfy additional *statistical* properties that preserve specific properties of the private data. For instance, fairness-relevant joint distributions whose distortion is a documented risk of DP synthesis, which may preserve or even amplify undesirable biases present in the original data [20]. To date, there is no solution tailored to repairing already generated DP synthetic data in a way that jointly addresses logical consistency and statistical fidelity.

A straightforward response to inconsistency or bias is to regenerate the synthetic data, but this approach is often prohibitively expensive due to the computational complexity of modern DP synthesis pipelines. An alternative is to apply existing data repair techniques [2, 23, 37, 50]. However, classical data repair focuses on enforcing integrity constraints and disregards the statistical objectives that motivated synthetic data generation in the first place, potentially yielding repaired datasets that diverge from the distribution of the private data on specific statistics.

The need for post-hoc repair arises in several concrete scenarios. **(S1) Late-discovered DCs:** Integrity constraints are often identified only after the synthetic data has already been published or distributed, e.g., domain experts may flag violations after the fact. Regeneration requires rerunning the synthesis mechanism, whereas repair is pure post-processing given released marginal targets and incurs additional privacy cost only when new targets must be acquired. **(S2) Computational cost:** State-of-the-art DP synthesizers can require hours of compute (see Section 6); repair is orders of magnitude faster. **(S3) Preserving the existing sample:** The released dataset may have been reviewed, annotated, or curated by data stewards; repair emphasizes retaining the original records, preserving the curated sample while fixing violations.

Motivated by these gaps, we propose *synthetic data repair*: given a synthetic dataset, a set of DCs, and noisy two-way marginals privately estimated from the true data, delete tuples so that the result satisfies the DCs while remaining faithful to the private data. DCs enforce logical validity; the marginals encode statistical fidelity to specific properties, e.g., fairness-relevant marginals such as (race, income), whose distortion under DP synthesis is a recognized concern [20], or ones that are badly approximated by the existing sample. For repair, we employ the most well-understood repair model, i.e., deletion-based repair [23, 36–38]. The relative error measures the normalized distance between a repaired marginal and its noisy private estimate. Instead of hard constraints, the repair

Table 1: DC violations and relative marginal error for synthetic vs. repaired data for different synthesizers on the Census dataset. Darker red indicates worse values; darker green indicates better values. Marginal error reports the average relative error over all selected marginals, measuring how closely the (repaired) synthetic marginals match the private estimates. (*) Kamino was run on only 5K tuples due to scalability issues (see Example 1).

Synthesizer	Synthetic Data		Repaired Synth. Data	
	DCs Viol.	Marg. Error	DCs Viol.	Marg. Error
AIM	3.198×10^8	7.985	0	0.247
MST	1.93×10^7	10.576	0	0.45
PATE-CTGAN	1.48×10^{10}	666.34	0	0.754
KAMINO*	0	1.523	0	0.986

objective minimizes a single weighted loss, a convex combination of the deletion fraction and the average relative marginal error, with $\alpha \in [0, 1]$ trading off data retention against marginal fidelity, which is always feasible and avoids over-deletion. Although repair under DCs is already computationally hard [37], we prove that synthetic data repair is NP-hard even in the absence of DCs.

At the core of our framework are two complementary repair algorithms that minimize the weighted objective $\mathcal{L}$ while enforcing the DCs. The first, an MILP extending cost-aware subset repairs [36], yields optimal repairs but incurs a substantial runtime overhead. The second is a scalable two-phase greedy algorithm over the conflict graph [9]: it first resolves all DC conflicts with the maximum-degree vertex-cover heuristic, and then repeatedly deletes the tuple whose removal most decreases $\mathcal{L}$, scaling to datasets far beyond the MILP's reach. Our repair algorithms are agnostic to the source of the marginals. When suitable marginals are not available, Section 5 describes one possible private acquisition procedure via the one-shot top-k mechanism [13] to select poorly represented marginals and the Gaussian mechanism [5] to release their noisy values.

We evaluate our framework on four real-world datasets, repairing synthetic data from multiple state-of-the-art private generators. This tests our approach on *organic* constraint violations and statistical distortions arising naturally from synthesis, rather than injected ones, under realistic conditions. The hybrid method guarantees DC satisfaction while producing repairs that remain faithful to the private data without excessive deletions, and scales to large data sizes—a compelling balance of integrity, statistical fidelity, and scalability for post-hoc repair of privately generated synthetic data.

Example 1. Consider the Census dataset [19], with sensitive attributes such as income, education, and nativity. The data satisfies several inherent DCs, e.g., that two individuals cannot share a nativity and have different citizenship. An analyst wishes to draw conclusions from a DP synthetic dataset of size 500,000, generated with privacy budget $\epsilon = 0.5$ by four state-of-the-art synthesizers: AIM [41], MST [40], PATE-CTGAN [57], and Kamino [21]. DC violations are known to reduce the reliability and usability of synthetic data [21]. The analyst also seeks to preserve fairness-relevant marginals, such as (race, income) and (education, income), which DP noise may distort [20]. We therefore apply the Gaussian Mechanism [5] under zCDP with budget $\rho = 1$ to privately estimate $k = 50$ randomly selected two-way marginals, yielding 50 distance-based marginals whose relative errors (between the repaired and private marginals) the repair objective $\mathcal{L}$ minimizes directly. Table 1 reports the DC violations and marginal error per synthesizer, averaged over 10 runs. All synthesizers incur large marginal errors, and all but Kamino also produce substantial numbers of DC violations. Kamino, which enforces DCs at generation time, produces no violations, but at a steep cost: due to severe scalability issues it was run on only 5K tuples, and when training the model on the full dataset and trying to sample 10K tuples it already exceeded our 12-hour timeout; moreover, repair still reduces its marginal error from 1.523 to 0.986. Our greedy repair reduces DC violations to 0 and significantly lowers marginal errors, deleting 1.2% of the tuples for MST, 1.3% for AIM, 3.4% for Kamino, and 94.9% for PATE-CTGAN (owing to its extreme violation rate). Thus, when regeneration is impractical, repair enforces DC compliance and substantially reduces marginal errors on the existing data.

Contributions. This work makes the following contributions:

- **Problem formulation (Section 3).** We formalize *synthetic data repair under DP*, the first framework for post-hoc repair of DP synthetic data that jointly enforces DCs and a set of marginals (e.g., fairness-relevant marginals). We cast it as minimizing a single weighted objective $\mathcal{L}$ trading off deletions against relative marginal error, and prove it NP-hard for restricted settings.
- **Repair algorithms (Section 4).** At the core of our framework are two complementary repair algorithms: an exact MILP that optimally minimizes $\mathcal{L}$, and a scalable greedy heuristic that resolves DC conflicts via vertex cover approximation and then deletes tuples along the steepest decrease of $\mathcal{L}$. Both algorithms are agnostic to the source of the marginal targets. Section 5 illustrates one possible DP mechanism for obtaining such targets via the one-shot top-k mechanism and the Gaussian Mechanism.
- **Experimental evaluation (Section 6).** On four real-world datasets and multiple state-of-the-art DP generators, we exercise the repair on *organic* synthesis violations and study its quality–scalability tradeoffs, showing that our hybrid repair balances integrity, statistical fidelity, and performance.

We review the necessary background for the paper in Section 2, discuss related work in Section 7, and conclude in Section 8.

2 PRELIMINARIES

We next review the necessary definitions from previous work. Notations are summarized in Table 2.

2.1 Databases, Repairs, and Marginals

We consider a single-relation schema $\mathcal{A}ttr = (A_1, \ldots, A_m)$, where each attribute A_i has a domain $Dom(A_i)$ of values. A database D over $\mathcal{A}ttr$ is a collection of tuples in $Dom(A_1) \times \cdots \times Dom(A_m)$, treated as distinct occurrences (so duplicates can be deleted individually). A database D' is a *subset* of D, denoted $D' \subseteq D$, if it is obtained from D by deleting zero or more tuples. We consider a synthetic database D_s generated, under Differential Privacy, from a private database D_p over the same attribute set $\mathcal{A}ttr$; both are assumed non-empty throughout, and n denotes the data size ($|D_s|$ or $|D_p|$, depending on the context).

Denial Constraints. Following previous work on related topics [9, 38, 50], we focus on Denial Constraints (DCs) on pairs of tuples. Given a database D over a schema $\mathcal{A}ttr$, a DC can be written

ID	Edu-num	Edu.	Occu.	Income	Age	Race
t_1	14	Undergrad	Manager	50K	52	AA
t_2	13	HS	Craft-repair	47K	48	SA
t_3	14	Undergrad	Farming	50K	25	C
t_4	22	Prof_school	Prof	62K	57	AA

(a) Private dataset D_p^E.

ID	Edu-num	Edu.	Occu.	Income	Age	Race
t_1	15	Undergrad	Manager	50K	55	AA
t_2	12	HS	Craft-repair	40K	45	SA
t_3	15	Undergrad	Farming	50K	22	C
t_4	22	Prof_school	Prof	60K	55	AA
t_5	14	Undergrad	Prof	30K	50	AA
t_6	16	Undergrad	Craft-repair	60K	50	AA

(b) Synthetic dataset D_s^E.

Figure 1: Private and synthetic databases based on population datasets [19, 33].

as $\forall t_1, t_2 \in D$, $\neg(\phi_1 \wedge \ldots \wedge \phi_p)$, where ϕ_i is a predicate of the form $t_i[A_j] \circ v$ or $t_1[A_j] \circ t_2[A_j]$, for $i \in \{1,2\}$, $A_j \in \mathcal{A}ttr$, $v \in Dom(A_j)$, and $\circ \in \{>,<,\geq,\leq,=,\neq\}$. DCs are a general form of integrity constraint, capturing multiple classes such as Functional Dependencies (FDs), Conditional FDs [4], and metric FDs [30]. If all tuple pairs in D satisfy a set of DCs Σ, we write $D \models \Sigma$.

Database Repairs. Following previous work [23, 36, 37], we consider repairs via tuple deletions: given a set of DCs Σ, a *consistent subset* of D_s is a database $D_r \subseteq D_s$ such that $D_r \models \Sigma$.

Example 2. Consider the databases D_p^E and D_s^E in Figure 1 based on an income dataset [19, 33]. The database schema is $\mathcal{A}ttr = \{ID, Edu-num, Edu., Occu., Income, Age, Race\}$ and it includes $n = 4$ tuples. Consider the following two DCs:

(1) $\sigma_1 : \forall t_1, t_2 \neg(t_1[Edu] = t_2[Edu] \wedge t_1[Edu-num] \neq t_2[Edu-num])$

(2) $\sigma_2 : \forall t_1, t_2 \neg(t_1[Occu] = \text{'Prof'} \wedge t_2[Occu] = \text{'Craft-repair'} \wedge t_1[Income] < t_2[Income])$

Notice that $D_p^E \models \sigma_1, \sigma_2$ while $D_s^E \not\models \sigma_1, \sigma_2$. Specifically, (t_1, t_5), (t_1, t_6), (t_3, t_5), (t_3, t_6), and (t_5, t_6) violate σ_1, and (t_2, t_5), (t_5, t_6) violate σ_2. A repair of D_s^E would be removing t_5 and t_6.

Marginals. We formalize the notion of 2-way marginals, drawing on the prior work for generating private synthetic data [40, 42, 58].

Definition 3 (2-way marginal). *Given a pair of attributes $A_i, A_j \in \mathcal{A}ttr$, and a dataset D over $\mathcal{A}ttr$, the 2-way marginal of $a_i \in Dom(A_i)$, $a_j \in Dom(A_j)$ in a non-empty D is a normalized count*

$$m_D(i,j) := \frac{1}{|D|} \sum_{t \in D} \mathbb{1}[t[A_i] = a_i \wedge t[A_j] = a_j]$$

Note that the normalization is always by $|D|$, the size of the dataset whose marginal is taken (e.g., $|D_r|$ for a repaired database). For the empty dataset we set $m_\emptyset(i,j) := 0$ by convention (all counts are zero).

Example 4. Reconsider D_p^E and D_s^E from Figure 1 (writing $Under$ for $Undergrad$). Let m_1 and m_2 be the 2-way marginals of the value pairs (Inc=50K, Edu=$Under$) and (Edu=$Under$, $Race$=AA). In D_p^E (n=4), m_1=$\frac{2}{4}$=0.5 and m_2=$\frac{1}{4}$=0.25; in D_s^E (n=6), m_1=$\frac{2}{6}$≈0.33 and m_2=$\frac{3}{6}$=0.5. Hence D_s^E under-represents m_1 and over-represents m_2 relative to D_p^E.

Table 2: Notation.

Symbol	Meaning
$\mathcal{A}ttr$	Attribute set
D_p	Private dataset
D_s	Synthetic dataset
D_r	Repaired dataset
$\sigma \in \Sigma$	Denial Constraint (DC)
$m_D(i,j)$	Marginal on D
u	Utility function for the exponential mechanism

2.2 Differential Privacy

We give the necessary DP background for the part of our framework that requires private data access to obtain marginals.

Definition 5 (Neighboring databases). *Two databases D and D' are considered neighboring databases, denoted by $D' \approx D$, if D and D' have the same size and differ by at most one tuple.*

We adopt *zero-concentrated differential privacy* (zCDP) [5] as our privacy notion, following common practice [40, 41]: it has a single parameter ρ that directly measures the privacy loss (smaller means more private), it yields tight bounds on the cumulative loss of multiple private releases, and it is satisfied by the Gaussian noise we employ. Standard conversions exist between ρ-zCDP and the classical (ϵ, δ)-DP guarantee [5, 16]; in particular, every zCDP mechanism also satisfies classical DP.

Definition 6 (Zero-Concentrated Differential Privacy (zCDP) [5]). *A mechanism $\mathcal{A}$ is said to be ρ-*zero-concentrated differential private, *or ρ-zCDP for short, if for any neighboring datasets D and D' and all $\alpha \in (1, \infty)$ it holds that*

$$D_\alpha(\mathcal{A}(D) \| \mathcal{A}(D')) \leq \rho\alpha$$

where $D_\alpha(\mathcal{A}(D) \| \mathcal{A}(D'))$ denotes the Rényi divergence of the distribution $\mathcal{A}(D)$ from the distribution $\mathcal{A}(D')$ at order α [46].

Intuitively, zCDP bounds the divergence between the output distributions on neighboring databases, so that no single tuple can influence the output too much. zCDP composes additively across sequential private releases:

Proposition 7 (Sequential composition for zCDP [5, 44]). *Let $\mathcal{A}_1 : \mathcal{D} \to \mathcal{Y}$ and $\mathcal{A}_2 : \mathcal{Y} \to \mathcal{Z}$. Suppose $\mathcal{A}_1$ satisfies ρ_1-zCDP, $\mathcal{A}_2(\cdot)$ satisfies ρ_2-zCDP. Define $\mathcal{A}_3 : \mathcal{D} \to \mathcal{Z}$ by $\mathcal{A}_3(D) = \mathcal{A}_2(\mathcal{A}_1(D))$. Then, $\mathcal{A}_3$ satisfies $(\rho_1 + \rho_2)$-zCDP.*

The Gaussian Mechanism. The Gaussian Mechanism is the standard building block for releasing real-valued (or vector-valued) queries under zCDP. It adds independent Gaussian noise calibrated to the query's ℓ_2 sensitivity and the desired privacy budget.

Definition 8 (ℓ_2 Sensitivity). *Given a function $f : \mathcal{D} \to \mathbb{R}^k$, its ℓ_2* sensitivity *is $\Delta_2 f = \max_{D' \approx D} \|f(D) - f(D')\|_2$.*

For a scalar function ($k = 1$) we write simply Δ_f (e.g., Δ_u for a utility function u).

Definition 9 (Gaussian Mechanism [5]). *Given a database D, a function $f : \mathcal{D} \to \mathbb{R}^k$ with ℓ_2 sensitivity $\Delta_2 f$, and a privacy parameter $\rho > 0$, the* Gaussian Mechanism *outputs*

$$\mathcal{M}_G(D) \;=\; f(D) + \mathcal{N}\big(0,\, \sigma^2 I_k\big), \qquad \sigma = \frac{\Delta_2 f}{\sqrt{2\rho}},$$

and satisfies ρ-zCDP.

The noise standard deviation σ trades off privacy (smaller ρ requires larger σ) against accuracy (larger σ means noisier output).

The Exponential Mechanism (EM). Given a set of candidates $\mathcal{R}$ and a utility function $u : \mathcal{R} \times \mathcal{D} \to \mathbb{R}$ with sensitivity Δ_u, the EM [43] selects a candidate $r \in \mathcal{R}$ with probability proportional to $\exp\big(\epsilon\, u(D, r)/(2\Delta_u)\big)$, thus favoring high-utility candidates. It satisfies ϵ-DP, and hence $\frac{1}{2}\epsilon^2$-zCDP [5].

3 SYNTHETIC DATA REPAIR

We now provide the required definitions for our framework and discuss the problem of synthetic data repair.

We first define a relative marginal error, as marginals capture important and relevant statistical properties that we aim for.

Definition 10 (2-way marginal relative error). *Let $\mathcal{A}ttr$ be a schema, $A_i, A_j \in \mathcal{A}ttr$ a pair of attributes, and $(a_i, a_j) \in Dom(A_i) \times Dom(A_j)$ a value pair. Let $\tilde{m}_{D_p}(A_i{=}a_i, A_j{=}a_j)$ be a noisy estimate of the private marginal (well-defined, as D_p is non-empty), assumed positive. The* 2-way marginal relative error (marginal) *for $(A_i{=}a_i, A_j{=}a_j)$ is*

$$E_m(m_{D_r}(A_i{=}a_i, A_j{=}a_j)) = \frac{|m_{D_r}(A_i{=}a_i, A_j{=}a_j) - \tilde{m}_{D_p}(A_i{=}a_i, A_j{=}a_j)|}{\tilde{m}_{D_p}(A_i{=}a_i, A_j{=}a_j)},$$

i.e., the distance between the repaired dataset's marginal and the noisy private estimate, normalized by the estimate. Unlike D_p and D_s, the repaired dataset D_r may be empty, in which case $m_{D_r}(A_i{=}a_i, A_j{=}a_j) = 0$ by Definition 3 and the relative error equals 1.[1]

Intuitively, the relative error, used by prior work [29, 31, 32], measures how faithfully the repaired marginal matches the noisy private estimate, normalized so that deviations are penalized proportionally to their target rather than in absolute terms. Rather than a hard constraint, it enters the repair objective as a soft term (Definition 11), keeping the problem always feasible.

For compact notation (see Table 2), we write a marginal as $E_m(m_{D_r}(i, j))$ and denote the set of monitored value pairs by $\mathcal{M}$.

We now define a synthetic data repair and an optimal synthetic data repair, drawing inspiration from work on data repair [36, 37].

Definition 11 (Synthetic data repair). *Given a database D, a set of marginals $\mathcal{M}$, and a set of DCs Σ, a synthetic data repair of D is a consistent subset $D_r \subseteq D$ satisfying $D_r \models \Sigma$. An* optimal *synthetic repair $D_r^* \subseteq D_s$ minimizes the weighted objective*

$$\mathcal{L}(D_r) = \alpha \cdot \frac{n - |D_r|}{n} + (1 - \alpha) \cdot \frac{1}{|\mathcal{M}|} \sum_{(A_i=a_i, A_j=a_j) \in \mathcal{M}} E_m(m_{D_r}(i, j))$$

where $n = |D_s| \geq 1$ (recall that D_s is non-empty) and $\alpha \in [0, 1]$ balances data retention against marginal fidelity.

Note that $\mathcal{L}$ is well-defined on every subset. In particular, for $D_r = \emptyset$, the convention $m_\emptyset(i, j) = 0$ (Definition 3) gives $E_m(m_\emptyset(i, j)) = 1$ for every monitored marginal, hence $\mathcal{L}(\emptyset) = \alpha + (1 - \alpha) = 1$ (and $\mathcal{L}(\emptyset) = \alpha$ when $\mathcal{M} = \emptyset$). This implies an interesting observation: a data repair that retains some of the tuples with high marginal error (larger than 1) will be inferior to an empty repair. In practice, our heuristic method (Section 4.2) always returns marginal errors smaller than 1, and thus finds a solution that retains tuples and in fact, incurs a relatively small deletion ratio.

We derive Definition 11 from the classic deletion-based data repair problem, modeled as an optimization problem [23, 37, 38], but incorporate 2-way marginals via a soft weighted objective $\mathcal{L}$ that jointly minimizes deletions and the relative marginal error. The second term averages the 2-way marginal relative errors of the monitored marginals (Definition 10). When $\mathcal{M} = \emptyset$, the second term is defined to be 0, and $\mathcal{L}$ reduces to the weighted deletion fraction $\alpha\,(n - |D_r|)/n$. The two terms are at different scales by design, prioritizing statistical fidelity. Among all consistent subsets, the optimal repair is the one that minimizes $\mathcal{L}$.

Finally, note that the set of provided marginals $\mathcal{M}$ does not need to contain all 2-way marginals of the private data.

Example 12. Reconsider the databases D_p^E and D_s^E in Figure 1 and the 2 DCs from Example 2. An example of marginals that hold in D_p^E are $E_m(m_{D_r}(Inc{=}50K, Edu{=}Under))$ (large in D_s^E, whose synthetic marginal deviates significantly from the private estimate) and $E_m(m_{D_r}(Edu{=}Under, Race{=}AA))$, where $Under$ is short for $Undergrad$. To get an optimal synthetic data repair of D_s^E we can remove t_5 and t_6: this satisfies both DCs (as shown in Example 2) and minimizes the weighted objective $\mathcal{L}$ from Definition 11, as removing t_5 and t_6 reduces both marginal relative errors while limiting deletions.

Problem definition. We now define the problem addressed in this paper to achieve a framework for synthetic data repair.

Definition 13 (Synthetic data repair). *Given a synthetic database D_s, a set of marginals $\mathcal{M}$, and a set of DCs Σ, find an optimal synthetic data repair for D_s.*

Some data repair approaches do attempt to preserve the statistics that exist in the data [50], albeit with an update-based repair model, rather than a deletion-based one. However, in our problem setting, we aim at *approximating specific statistical properties that may even differ from the current database statistics*, as we emphasize statistics originating in the private data, rather than to the synthetic data. Indeed, recall Example 1 where the synthetic data had a significant deviation from the specific marginals we mentioned.

We aim to provide a repair with a small distance from the initial synthetic database. However, finding the optimal synthetic data repair is NP-hard, as we now show.

Hardness of optimal synthetic data repair. Interestingly, the hardness of our problem requires neither both ingredients of $\mathcal{L}$ nor multiple constraints of either type: the marginal terms alone suffice without any DCs (via a reduction from Max-Cut [26]), a single DC alone suffices even with no marginals (via a reduction from Independent Set [26]), and so does the minimal combination of one DC and one marginal. Here the schema, dataset, and marginals are all part of the input.

Proposition 14 (Hardness of Definition 13). *The decision version of Definition 13 is NP-hard in each of the following settings:*

(1) $\Sigma = \emptyset$, with the marginal terms of $\mathcal{L}$ alone;
(2) $|\Sigma| = 1$ and $|\mathcal{M}| = 1$;

[1] If we perform a noisy estimate of $\tilde{m}_{D_p}(A_i{=}a_i, A_j{=}a_j)$ and get 0, we can floor the denominator at a small $\tau > 0$ (e.g., $\tau = 1/|D_p|$ where $|D_p|$ is noisy as well).

(3) $|\Sigma| = 1$ *and* $\mathcal{M} = \emptyset$*, i.e., a single DC and no statistical constraints.*

We defer the proofs to Section A due to space limitations.

We begin by tackling the repair problem, assuming that the marginals are available (Section 4), and illustrate a possible way of obtaining them in the next section (Section 5).

4 COMPUTING SYNTHETIC DATA REPAIR

We next present two approaches for solving Definition 13: (1) an MILP formulation of the problem that yields an exact solution, and (2) a heuristic two step approach relying on first removing all the violations in the data and then improving the marginals.

4.1 An MILP Formulation

We next detail an MILP formulation of the problem. Each binary variable x_l represents a tuple $t_l \in D_s$: t_l is retained by the repair iff $x_l = 1$ in the solution to the program. The MILP minimizes the weighted objective $\mathcal{L}$ of Definition 11, a convex combination of the deletion fraction and the average relative marginal error (Definition 10), and we derive it in three steps. Throughout, $n = |D_s|$ is fixed, and for a marginal $m = (A_i{=}a_1, A_j{=}a_2)$ we write $C_m(x)$ for its retained count, so that the repaired marginal is

$$m_{D_r}(A_i{=}a_1, A_j{=}a_2) = \frac{C_m(x)}{\sum_l x_l}, \quad C_m(x) = \sum_{\substack{l:\, t_l[A_i]=a_1,\\ t_l[A_j]=a_2}} x_l, \qquad (1)$$

normalized by the *variable* repair size $\sum_l x_l$ (for a non-empty repair; the empty repair is handled in Step 3).

Step 1: capturing the relative errors. For each $m \in \mathcal{M}$ we introduce a continuous variable $d_m \geq 0$ whose intended value is the marginal distance of the repair,

$$d_m \;=\; \left| m_{D_r}(A_i{=}a_1, A_j{=}a_2) - \tilde{m}_{D_p}(a_1, a_2) \right|. \qquad (2)$$

In the objective, d_m is divided by its private estimate $\tilde{m}_{D_p}(a_1, a_2)$ (a positive constant), so the second objective term equals the average relative marginal error $E_m(m_{D_r}(A_i{=}a_1, A_j{=}a_2))$ of Definition 10.

Step 2: exact linearization. Enforcing Equation (2) directly is nonlinear, since by Equation (1) the repaired marginal divides by the variable repair size. Multiplying Equation (2) through by $\sum_l x_l$ yields the pair of constraints

$$d_m \textstyle\sum_l x_l \;\geq\; \pm\big(C_m(x) - \tilde{m}_{D_p}(a_1, a_2) \sum_l x_l\big), \qquad (3)$$

whose only remaining nonlinearity is the bilinear term $d_m \sum_l x_l$. We linearize it *exactly*: for each marginal m and tuple t_l, a product variable $z_{ml} = d_m \cdot x_l$ is encoded by the standard constraints $z_{ml} \leq d_m$, $z_{ml} \leq x_l$, $z_{ml} \geq d_m - (1 - x_l)$, and $z_{ml} \geq 0$. These are exact because x_l is binary: they force $z_{ml} = d_m$ when $x_l = 1$ and $z_{ml} = 0$ when $x_l = 0$ (in the latter case they also imply $d_m \leq 1$, which excludes no intended solution, since marginals lie in $[0, 1]$). Substituting $\sum_l z_{ml}$ for $d_m \sum_l x_l$ in Equation (3) makes every constraint linear, so the program is a genuine MILP.

Step 3: DCs and the empty repair. DC conflicts are resolved as before with $x_i + x_j \leq 1$ for every conflicting pair t_i, t_j. The empty repair requires no special treatment: by the convention of Definition 3 its marginals are 0, so each relative error equals 1 and $\mathcal{L}(\emptyset) = 1$; the safeguard constraints $d_m \geq \tilde{m}_{D_p}(a_1, a_2)(1 - \sum_l x_l)$, inactive whenever at least one tuple is retained, force the MILP objective to this exact value at the all-zeros point. Since the marginals appear only in the objective (not as hard constraints), the MILP is always feasible. The complete program is shown in Equation (4). The correctness of the program is established by the following proposition (proof in Section B).

Proposition 15. *Given a synthetic database D_s, a set of DCs Σ, and a set of marginals $\mathcal{M}$, an optimal solution to the MILP in Equation* (4) *induces an optimal synthetic data repair, i.e., the repair $D_r = \{t_l \mid x_l = 1\}$ minimizes $\mathcal{L}(D_r) = \alpha \cdot (n - |D_r|)/n + (1 - \alpha) \cdot (1/|\mathcal{M}|) \cdot \sum_{m \in \mathcal{M}} E_m(m_{D_r}(i, j))$ over all DC-consistent subsets of D_s.*

$$\begin{aligned}
\mathbf{min}\ \ & \alpha \cdot \frac{n - \sum_{l=1}^{n} x_l}{n} + \frac{1-\alpha}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \frac{d_m}{\tilde{m}_{D_p}(a_1, a_2)} \qquad (4)\\
\mathbf{s.t.}\ \ & x_i + x_j \leq 1, \quad \forall\, t_i, t_j \text{ conflicting on } \Sigma\\
& \textstyle\sum_l z_{ml} \geq \sum_{\substack{l:\, t_l[A_i]=a_1,\\ t_l[A_j]=a_2}} x_l \;-\; \tilde{m}_{D_p}(a_1, a_2) \sum_l x_l, \ \forall m \in \mathcal{M}\\
& \textstyle\sum_l z_{ml} \geq \tilde{m}_{D_p}(a_1, a_2) \sum_l x_l \;-\sum_{\substack{l:\, t_l[A_i]=a_1,\\ t_l[A_j]=a_2}} x_l, \ \forall m \in \mathcal{M}\\
& z_{ml} \leq d_m, \quad z_{ml} \leq x_l, \quad z_{ml} \geq d_m - (1 - x_l), \ \forall m, l\\
& d_m \geq \tilde{m}_{D_p}(a_1, a_2)\big(1 - \textstyle\sum_l x_l\big), \ \forall m \in \mathcal{M}\\
& d_m \geq 0, \ \ z_{ml} \geq 0, \ \ x_l \in \{0, 1\}, \quad \forall\, m \in \mathcal{M}, \ l \in [1, n]
\end{aligned}$$

Example 16. Consider the databases in Figure 1b, the two DCs from Example 2, and the two marginals from Example 12. The corresponding MILP contains the DC-expressing constraints $x_1 + x_5 \leq 1$, $x_1 + x_6 \leq 1$, $x_2 + x_5 \leq 1$,$x_3 + x_5 \leq 1$, $x_3 + x_6 \leq 1$, $x_5 + x_6 \leq 1$. For the marginal $m_1 = (Inc{=}50K, Edu{=}Under)$ (with noisy estimate $\tilde{m}_{D_p}(50K, Under)$ and $n = 6$), it contains the two distance constraints $\sum_{l=1}^{6} z_{m_1 l} \geq \sum_{l: t_l[Inc]=50K, t_l[Edu]=Under} x_l - \tilde{m}_{D_p}(50K, Under) \sum_{l=1}^{6} x_l$ and $\sum_{l=1}^{6} z_{m_1 l} \geq \tilde{m}_{D_p}(50K, Under) \sum_{l=1}^{6} x_l - \sum_{l: t_l[Inc]=50K, t_l[Edu]=Under} x_l$, together with the linearization $z_{m_1 l} = d_{m_1} x_l$ for $l \in [1, 6]$ (encoded by $z_{m_1 l} \leq d_{m_1}$, $z_{m_1 l} \leq x_l$, $z_{m_1 l} \geq d_{m_1} - (1 - x_l)$, $z_{m_1 l} \geq 0$), and analogously for $m_2 = (Edu{=}Under, Race{=}AA)$. The objective is $\alpha \cdot (6 - \sum_l x_l)/6 + (1-\alpha)/2 \cdot \big(d_{m_1}/\tilde{m}_{D_p}(50K, Under) + d_{m_2}/\tilde{m}_{D_p}(Under, AA)\big)$, where each distance d_m is normalized by its private estimate.

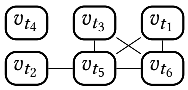
Note that the size of the MILP can be large: beyond the $|\mathcal{M}|$ distance variables d_m (with $|\mathcal{M}| \leq \binom{|\mathcal{A}ttr|}{2} \cdot \max_{A \in \mathcal{A}ttr} |Dom(A)|^2$), the exact linearization introduces $|\mathcal{M}| \cdot n$ product variables z_{ml} and $O(|\mathcal{M}| \cdot n)$ associated constraints. Together with the $O(|D|^2)$ conflict constraints, the program size is $O(|D|^2 + |\mathcal{M}| \cdot |D|)$.

Recall that the MILP is always feasible (Step 3); in particular, when $D_s \models \Sigma$, retaining all tuples is feasible, so the program applies even when the synthetic data merely exhibits marginal deviations, with α selecting the deletions–fidelity trade-off. As a result, we show that it encounters scalability issues as well as a large deletion ratio to get the exact optimum (see Section 6). This motivates us to explore a heuristic solution.

4.2 Hybrid: A Two-Step Heuristic

In classic data repair, the objective is to satisfy integrity constraints while maximizing the number of retained tuples. This problem can be reduced to *Vertex Cover* (VC) by constructing a conflict graph [9, 37]. Given a dataset D_s and constraints Σ, the conflict graph $G = (V, E)$ contains one vertex per tuple and an edge between two vertices whenever the corresponding tuples jointly violate a constraint. A vertex cover thus corresponds to a set of tuples whose removal eliminates all violations [38].

Example 17. The DCs in Example 2 induce the following conflict graph for D_s^E from Figure 1b:



Our heuristic (Algorithm 1) addresses the two objectives of Definition 11 in two phases. The first phase deletes tuples until the data is DC-consistent, and the second greedily deletes additional tuples to reduce the distance to the private marginals.

Phase 1: Resolving conflicts. We first eliminate all DC violations using the standard greedy VC heuristic: while the conflict graph contains edges, we remove the vertex of maximum degree (breaking ties arbitrarily) together with its incident edges (lines 2–5). Removing the highest-degree vertex resolves the largest number of violations per deletion, keeping the repair small. Since DCs are anti-monotone, deleting tuples never introduces new violations, so the phase terminates with a DC-consistent dataset.

Phase 2: Reducing the repair loss. The first phase ignores statistical fidelity. The second phase greedily deletes tuples to reduce the repair loss $\mathcal{L}$ of Definition 11. Starting from the DC-consistent dataset, at each iteration the algorithm removes the tuple whose deletion yields the largest decrease in $\mathcal{L}$ (lines 6–8), stopping when no single deletion reduces it further. Because Phase 2 only deletes tuples, it preserves the DC-consistency established in Phase 1. As an implementation optimization of this phase, we perform batch removal instead of deleting a single tuple per iteration. Rather than immediately deleting the tuple with the smallest individual loss, we first identify the k tuples with the lowest hypothetical removal loss and then determine the largest removable prefix of this sorted list. The resulting batch is deleted in a single iteration, substantially reducing the number of expensive evaluations of $\mathcal{L}$. The complete procedure is given in Section D due to space limitations.

The loss function is efficiently computed for each tuple using prior results (details in Section C due to space limitations).

Example 18. We trace Algorithm 1 on D_s^E (Figure 1b, $n = 6$), using the conflict graph of Example 17 and the two marginals of Example 12, whose noisy private estimates are $\tilde{m}_{D_p}(Inc{=}50K, Edu{=}Under) = 0.5$ and $\tilde{m}_{D_p}(Edu{=}Under, Race{=}AA) = 0.25$.
Phase 1 (conflict resolution). The initial degrees are $\deg(v_{t_5}) = 4$, $\deg(v_{t_6}) = 3$, $\deg(v_{t_1}) = \deg(v_{t_3}) = 2$, and $\deg(v_{t_2}) = 1$, while v_{t_4} is isolated. The algorithm removes the maximum-degree vertex v_{t_5} together with its incident edges. The remaining edges are $\{v_{t_1}, v_{t_6}\}$ and $\{v_{t_3}, v_{t_6}\}$, so v_{t_6} now has the largest degree (2) and is removed next, leaving no edges. Phase 1 deletes $\{t_5, t_6\}$ and produces the DC-consistent dataset $D_r = \{t_1, t_2, t_3, t_4\}$.
Phase 2 (loss minimization). Phase 2 minimizes the loss $\mathcal{L}$ of Definition 11 with $\alpha = 0.5$ (here $n = |D_s^E| = 6$). In D_r ($|D_r| = 4$), the monitored marginals are $m_{D_r}(Inc{=}50K, Edu{=}Under) = \frac{2}{4} = 0.5$ (from t_1, t_3) and $m_{D_r}(Edu{=}Under, Race{=}AA) = \frac{1}{4} = 0.25$ (from t_1), which

Algorithm 1: HYBRID: Two-Step Synthetic Data Repair

input : D_s, conflict graph $G = (V, E)$ based on D_s and Σ, α, and a set of marginals $\mathcal{M}$.
output: Repaired dataset D_r.

```
1 D_r ← D_s;
  /* Phase 1: resolve DC conflicts via greedy VC.     */
2 while E ≠ ∅ do
3   v* ← arg max_{v∈V} deg(v);
4   D_r ← D_r \ {t_{v*}};
5   Delete v*, its incident edges, and any resulting isolated
      vertices from G; update V, E;
  /* Phase 2: greedily reduce the loss ℒ
     (Definition 11).                                 */
6 while ∃ t ∈ D_r : ℒ(D_r \ {t}) < ℒ(D_r) do
7   t* ← arg min_{t∈D_r} ℒ(D_r \ {t});
8   D_r ← D_r \ {t*};
9 return D_r;
```

match their private estimates, so $E_m(m_{D_r}(Inc{=}50K, Edu{=}Under)) = E_m(m_{D_r}(Edu{=}Under, Race{=}AA)) = 0$ and $\mathcal{L}(D_r) = 0.5 \cdot \frac{6-4}{6} + 0.5 \cdot 0 = \frac{1}{6}$. The algorithm then evaluates the deletion of each remaining tuple; a deletion drops $|D_r|$ to 3 (deletion fraction $\frac{6-3}{6} = 0.5$) and shifts the two marginals (m_1 over $(50K, Under)$, m_2 over $(Under, AA)$), yielding (with avg. E_m the mean of the two marginals' relative errors):

remove	m_1	m_2	avg. E_m	$\mathcal{L}(D_r \setminus \{t\})$
t_1	1/3	0	0.67	0.58
t_2	2/3	1/3	0.33	0.42
t_3	1/3	1/3	0.33	0.42
t_4	2/3	1/3	0.33	0.42

Every candidate deletion strictly increases $\mathcal{L}$ above $\frac{1}{6}$ (the added deletion cost is not offset by any marginal gain, since the marginals already match their targets), so the loop condition in line 6 fails and the algorithm terminates, returning $D_r = \{t_1, t_2, t_3, t_4\}$. This coincides with the optimal repair of Example 12.

Alternative heuristics that do not work. We examined three reasonable alternatives to Algorithm 1. Replacing Phase 1 with an *optimal* DC repair (or any DC-only algorithm) barely reduces deletions over the maximum-degree heuristic yet is NP-hard and does not scale (Section 6). Replacing Phase 2 with the MILP of Proposition 15 lowers marginal error only slightly at a prohibitive cost: on Census data from AIM it improves the error from 0.23 to 0.09 at nearly identical deletions ($\approx 1.3\%$), but takes over 6 hours versus roughly 8 seconds for HYBRID. Collapsing both phases into a single *weighted* vertex cover performs almost identically to Phase 1 alone, minimizing a cover's weight does not minimize the final loss, and cannot operate when the conflict graph has no edges.

Complexity analysis. The complexity of Algorithm 1 is $O(|\mathcal{M}|n^2)$, where $n = |V|$. Phase 1 performs at most n iterations, each removing a single vertex, for $O(n^2)$ in total. In Phase 2, the effect of a deletion on $\mathcal{L}$ can be evaluated incrementally in $O(|\mathcal{M}|)$ per tuple (details in Section C), so each of the at most n iterations costs $O(|\mathcal{M}|n)$, giving $O(|\mathcal{M}|n^2)$ overall.

5 AN ILLUSTRATIVE PRIVATE MECHANISM FOR MARGINAL ACQUISITION

The repair problem in Definition 13 treats the monitored marginals $\mathcal{M}$ and their noisy target values as inputs. Our repair algorithms are agnostic to how these targets are selected or released. In practice, they may be specified by an analyst (e.g., fairness-relevant marginals), obtained from statistics already released by the synthesizer, or produced by any suitable DP query-selection mechanism. For completeness, this section describes one possible DP procedure for obtaining such targets when the data owner retains access to the private dataset. The procedure prioritizes two-way marginals that differ substantially between the private and synthetic datasets, using one-shot top-k selection followed by Gaussian measurement. We present it to demonstrate that marginal targets compatible with our repair formulation can be obtained with formal DP guarantees.

An illustrative mechanism. The algorithm (pseudocode in Section E.1) gets two privacy parameters, one for top-k selection, ρ_{Topk}, and one for noisily estimating them, ρ_{marg}. It proceeds in two stages. In the *selection* stage, it computes all candidate 2-way marginals of D_p and D_s and scores each candidate by the utility function u (Definition 20), perturbed with Gumbel noise of scale $2\Delta_u\sqrt{k/(8\rho_{Topk})}$, where Δ_u is the sensitivity of u; the k candidates with the highest noisy scores are selected. This one-shot top-k selection [13, 14, 49] is equivalent to k sequential Exponential Mechanisms, satisfies ρ_{Topk}-zCDP [6, 12–14, 51], and avoids a linear privacy blowup in the number of candidates. In the *measurement* stage, each selected marginal is released as a vector f of all its entries via the Gaussian Mechanism (Definition 9): independent noise $\mathcal{N}(0, \sigma^2_{\text{marg}})$ is added to every entry, with $\sigma_{\text{marg}} = \Delta_2 f/\sqrt{2\rho_{marg}/k} = \sqrt{k}/(n\sqrt{\rho_{marg}})$ calibrated to the per-marginal budget ρ_{marg}/k and to the ℓ_2 sensitivity (Definition 8) $\Delta_2 f = \sqrt{2}/n$, where $n = |D_p|$ (replacing one tuple of D_p changes at most two entries by $1/n$); the noisy entries are then clipped to $[0, 1]$ and renormalized. The output $\mathcal{M}$ consists of the value-pair entries of the selected marginals with their noisy estimates. We thus get the following result, slightly abusing notation to denote by $\mathcal{M}_0$ the set of all marginals (proof in Section E).

Proposition 19 (Guarantees for the illustrative procedure). *Given the utility function u with sensitivity Δ_u, a set of marginals $\mathcal{M}_0$, privacy parameters $(\rho_{Topk}, \rho_{marg})$, and a natural number k, the following holds:*

1. *The described procedure satisfies $(\rho_{marg} + \rho_{Topk})$-zCDP.*
2. *Denote by $OPT^{(i)}$ the i-th highest (true) score, and by $\mathcal{M}_0^{(i)}$ the i-th noisy private marginal selected by the one-shot top-k mechanism. For $\forall t$ and $\forall i \in \{1, 2, \dots, k\}$, we have*

$$Pr[u(\mathcal{M}_0^{(i)}) \leq OPT^{(i)} - \frac{2\Delta_u}{\sqrt{8\rho_{Topk}/k}}(\ln(|\mathcal{M}_0|) + t)] \leq e^{-t}$$

Utility function and attribute focus. Our utility function prioritizes the marginals most poorly represented in the synthetic data; The above procedure accepts any utility function.

Definition 20 (2-way marginals utility). *Given a private database D_p, a synthetic database D_s over $\mathcal{A}ttr$, and two attributes $A_i, A_j \in \mathcal{A}ttr$, the utility of the marginal on (A_i, A_j) is the ℓ_1 distance between its tables in the two databases:*

$$u(A_i, A_j) = \sum_{(a_i, a_j)} \left| m_{D_p}(a_i, a_j) - m_{D_s}(a_i, a_j) \right|,$$

where the sum ranges over $(a_i, a_j) \in Dom(A_i) \times Dom(A_j)$.

The sensitivity of the function is low, allowing us to compute it with relative accuracy under DP (proof in Section E).

Proposition 21. *The sensitivity of u is $\Delta_u = \frac{2}{n}$.*

When users prefer a specific subset of attributes, the algorithm can restrict its candidates to that subset (as in Example 1).

Example 22. Consider D_p^E and D_s^E (Figure 1) with k=2. By Example 4, (Edu=$Under$, $Race$=AA) deviates by $|0.25 - 0.5| = 0.25$ and (Inc=50K, Edu=$Under$) by only ≈ 0.17, so the former scores higher (up to Gumbel noise). Both marginals are selected, and the Gaussian Mechanism releases them with noise σ_{marg}, giving estimates $\tilde{m}_{D_p}(Edu{=}Under, Race{=}AA) \approx 0.25$ and $\tilde{m}_{D_p}(Inc{=}50K, Edu{=}Under) \approx 0.5$, i.e., the targets that drive the repair in Example 12.

The construction above establishes that the marginal targets consumed by our repair algorithms can, when necessary, be selected and measured under DP. This is only one possible acquisition strategy. Applications may prioritize other approaches such as domain-specified marginals, fairness-sensitive statistics, workload-relevant queries, or measurements already produced during synthesis.

6 EXPERIMENTS

We perform an experimental evaluation of our repair approach, aimed at answering the following questions: (1) How does our approach perform in terms of repair cost and marginal error compared to existing baselines? (2) How does our approach affect the faithfulness of the synthetic data to the private data compared to existing baselines? (3) How does our approach scale compared to the baselines? (4) What is the effect of the tradeoff parameter α on the performance of Algorithm 1?

Summary of our findings. Hybrid (Algorithm 1) eliminates *all* DC violations, up to 9.5×10^{10} violations, for every dataset and generator, and reduces the error on the targeted marginals in 15 of 16 configurations, by up to four orders of magnitude, while typically deleting under 10% of the tuples (for AIM and MST); global fidelity deteriorates by at most 24% (two-way TVD), and downstream ML accuracy often stays within ~1% (Section 6.2). Relative to the baselines, it trades slightly more deletions for substantially lower marginal error at a moderate runtime cost, whereas the exact MILP exceeds our twelve-hour limit on large instances (Section 6.3). The improvements on the noisy targets transfer to the *true* private marginals (Section 6.4), and performance is robust to the choice of the trade-off parameter α (Section 6.6).

6.1 Experimental Setup

All experiments were conducted on a machine with 8 CPU cores and 64 GB RAM using Python 3.11. We used Pandas and DuckDB for data processing and Gurobi to solve the MILPs.[2] To avoid a memory issue, we used a self-implementation of the conflict graph. Unless stated otherwise, each experiment was repeated using 10

[2] Source code: https://anonymous.4open.science/r/SyntheticDataRepair-5625

independently generated synthetic datasets and 10 independently sampled noisy marginal sets, resulting in 100 runs per configuration.

Datasets. We evaluate on four real-world datasets commonly used in data cleaning and synthetic data generation.

- **IPUMS-CPS (Census)** [19]: 3,395,604 tuples and 8 attributes (6 categorical and 2 bucketized numerical attributes) from the 2022 U.S. Census Population Survey.
- **Adult** [33]: 48,842 tuples and 15 demographic attributes derived from the U.S. Census.
- **COMPAS** [48]: 60,798 tuples and 17 attributes describing criminal history and recidivism.
- **Tax** [8]: 999,910 tuples and 9 demographic and tax-related attributes.

Prior to all experiments, numerical attributes were bucketized and identifier or near-identifier attributes (e.g., IDs) were removed.

Marginal targets. We evaluate repair conditional on a supplied set of statistical targets. Unless stated otherwise, each run - uses 50 two-way marginals sampled uniformly, and measured using the Gaussian mechanism with total budget $\rho = 1$ (recall that pure ϵ-DP implies $\epsilon^2/2$-zCDP). We sample each target set once and reuse the same noisy measurements across all methods. We generate 10 independent sets using different seeds and repeat every experiment on each set; each represents a separate $\rho = 1$ execution rather than a joint release. When varying the number of marginals, we use nested prefixes of the same ordered sample and keep the total budget fixed at $\rho = 1$, isolating the effect of additional targets from sampling variability and privacy expenditure. Random targets also decouple repair quality from target selection.

Denial constraints. Most denial constraints are taken from previous work [36, 47, 50]. For the sake of the experiments, additional constraints were identified after bucketization introduced new deterministic relationships between attributes. The complete constraint set is given in Section F.

Synthetic data generators. We evaluate our repair methods on synthetic datasets generated using state-of-the-art DP generators, including PATE-CTGAN [52], AIM [41], MST [40], and Kamino [21]. Evaluating on the output of these generators lets us test our approach on data with *organic* constraint violations and statistical distortions, those arising naturally from the synthesis process, exercising the system under realistic conditions. Unless stated otherwise, all generators use their default hyperparameters, privacy budget $\epsilon = 0.5$, and generate datasets with 500K tuples. As Kamino was not scalable to support large datasets, it was trained on a sample of 5K tuples of the data and generated synthetic data with 5K tuples.

Repair methods and baselines. We evaluate the proposed Hybrid repair algorithm (Algorithm 1) with a batch size of 100 tuples for deletion in Phase 2 and $\alpha = 0.5$; as Section 6.6 shows, repair quality is robust to the choice of α, varying only modestly across its range. We have compared Hybrid to the MILP (Section 4.1) in Section 6.5 showing that the MILP approach is infeasible in most cases. Therefore, we focus on examining the performance of Hybrid in this section. We compare against repair methods designed for data repairing based on DCs:

- **Optimal-DC-Baseline** [38]: the optimal ILP minimizing tuple deletions while ignoring marginal information.
- **Classic-VC-Baseline** [55]: the standard 2-approximation obtained by reducing data repair to the vertex cover problem on the conflict graph.
- **Vanilla-VC-Baseline** [10]: the unweighted greedy vertex cover algorithm, which repeatedly removes the tuple participating in the largest number of violations.

Evaluation metrics. We evaluate both repair cost and data quality using four metrics.

- **Deletion ratio:** The fraction of tuples removed during repair (first component in the loss function in Definition 11).
- **Relative marginal error:** As defined in Definition 10.
- **Two-way TVD:** The average Total Variation Distance (TVD) over all 2-way marginals between the repaired and private datasets.
- **ML accuracy:** Classification accuracy defined as $\frac{TP+TN}{TP+FP+FN+TN}$ on the private data using Logistic Regression, Random Forest, and MLP models trained on the repaired synthetic data.

6.2 Overall Evaluation

We give an overview evaluation of Algorithm 1: each generator produces a synthetic dataset under its default configuration, which we then repair. Table 3 summarizes repair quality along four axes: DC violations, marginal error, deletions, and global fidelity. Table 4 reports downstream ML utility.

Repair is necessary even alongside constraint-aware generation. The first column group of Table 3 shows that AIM, MST, and PATE-CTGAN produce between 10^7 and over 10^{10} violating pairs on every dataset. Kamino, which enforces DCs at generation time, attains zero violations on three of the four datasets but still produces 1.56×10^5 on Tax, and, unlike post-hoc repair, requires the DCs *before* synthesis (Scenario (S1) in Section 1); it also ran on only 5K tuples due to severe scalability issues, timing out at 12 hours even for 10K tuples. In every configuration, Algorithm 1 reduces the violation count to zero.

Consistency comes with a marked improvement in the target statistics and, in most settings, a modest deletion cost. The second column group reports the ratio of mean marginal error after repair to before (below one indicates improvement): repair lowers it in 15 of 16 configurations, often by several orders of magnitude; the sole exception is Tax under PATE-CTGAN, where the error rises by roughly 44.5%. Repair improves even Kamino's marginals (ratios of 0.81–0.88), showing that generation-time DC enforcement does not substitute for statistical repair. For deletions (third group), AIM and MST lose a small fraction of tuples, typically below 10% and never more than 30%, whereas PATE-CTGAN requires deleting almost the entire dataset; as Section 6.3 shows, this is not an artifact of our algorithm but a consequence of the extreme violation counts, and deletion-minimizing classical repairs behave similarly. For Kamino on Tax, the 55% deletion ratio is driven primarily by the loss-reduction phase rather than conflict resolution, consistent with its comparatively small violation count.

Beyond the targeted marginals, repair largely preserves global statistical structure. The final column group reports the ratio of two-way TVD to the private dataset after versus before repair. Since Algorithm 1 optimizes only the monitored marginals, tightening them can perturb the remaining distributions; the resulting TVD ratios grow by at most 24%, and in one configuration (Compas,

Table 3: Overall evaluation of Hybrid. The table reports the number of denial constraint (DC) violations before and after repair, the ratio of mean marginal error after repair (repaired/synthetic), the deletion ratio, and the ratio of two-way TVD after repair (repaired/synthetic). For the ratio columns, values below one indicate improvement, while values above one indicate deterioration. (*) Kamino was run on only 5K tuples due to severe scalability issues.

Dataset	DC Violations					Marginal Error Ratio (Repaired / Synthetic)				Deletion Ratio				Two-way TVD Ratio (Repaired / Synthetic)			
	AIM	KAMINO*	MST	PATE CTGAN	After Repair	AIM	KAMINO*	MST	PATE CTGAN	AIM	KAMINO*	MST	PATE CTGAN	AIM	KAMINO*	MST	PATE CTGAN
Adult	1.19×10^{10}	0	6.21×10^{9}	7.18×10^{10}	0	0.0011	0.8075	0.0013	0.0005	0.081	0.0111	0.069	0.998	1.0002	1.133	1.053	1.113
Census	3.20×10^{8}	0	1.93×10^{7}	1.48×10^{10}	0	0.195	0.839	0.414	0.0078	0.013	0.034	0.012	0.949	1.123	1.008	1.052	1.052
Compas	3.92×10^{8}	0	7.78×10^{8}	9.51×10^{10}	0	0.0024	0.881	0.172	2.2×10^{-5}	0.029	0.0055	0.034	0.975	1.064	1.0001	1.067	0.958
Tax	1.20×10^{9}	1.56×10^{5}	7.21×10^{8}	4.76×10^{9}	0	0.984	0.883	0.701	1.445	0.295	0.5479	0.202	0.990	1.241	1.0062	1.244	1.086

Table 4: Ratio of downstream classification accuracy after repair (repaired/synthetic). Values above one indicate improved predictive performance, values below one indicate deterioration, and one indicates no change.

Dataset	AIM			MST			PATE-CTGAN			Kamino		
	LR	RF	MLP	LR	RF	MLP	LR	RF	MLP	LR	RF	MLP
Adult	1.000	1.000	1.000	1.000	1.000	1.000	0.784	0.928	0.958	1.010	0.990	1.003
Census	1.002	1.000	1.000	0.998	1.001	1.010	0.487	1.115	1.059	1.000	1.000	1.000
Compas	1.000	1.000	1.000	1.000	1.000	1.000	0.640	1.022	0.715	1.021	1.091	0.999
Tax	1.000	1.000	1.000	1.000	1.000	1.000	0.853	1.187	0.794	1.170	1.330	2.346

PATE-CTGAN) global fidelity even improves. Overall, repair substantially improves the target statistics while largely retaining the global statistical properties of the synthetic data.

Finally, Table 4 compares downstream predictive performance by training Logistic Regression (LR), Random Forest (RF), and Multi-Layer Perceptron (MLP) classifiers on the repaired and synthetic data and evaluating them on the private data. For AIM and MST, repair leaves accuracy essentially unchanged (ratios between 0.998 and 1.010), showing that enforcing DCs has negligible impact on downstream utility. For PATE-CTGAN, there is mixed behavior: while repair substantially improves some models (up to 1.187×). It also causes noticeable degradation in others (down to 0.487×), reflecting the poor quality of the generated data and the difficulty of repairing it. For Kamino, repair consistently preserves or improves downstream performance, with gains reaching 2.346×, indicating that repairing higher-quality synthetic data can maintain constraint satisfaction while preserving, and occasionally enhancing, accuracy.

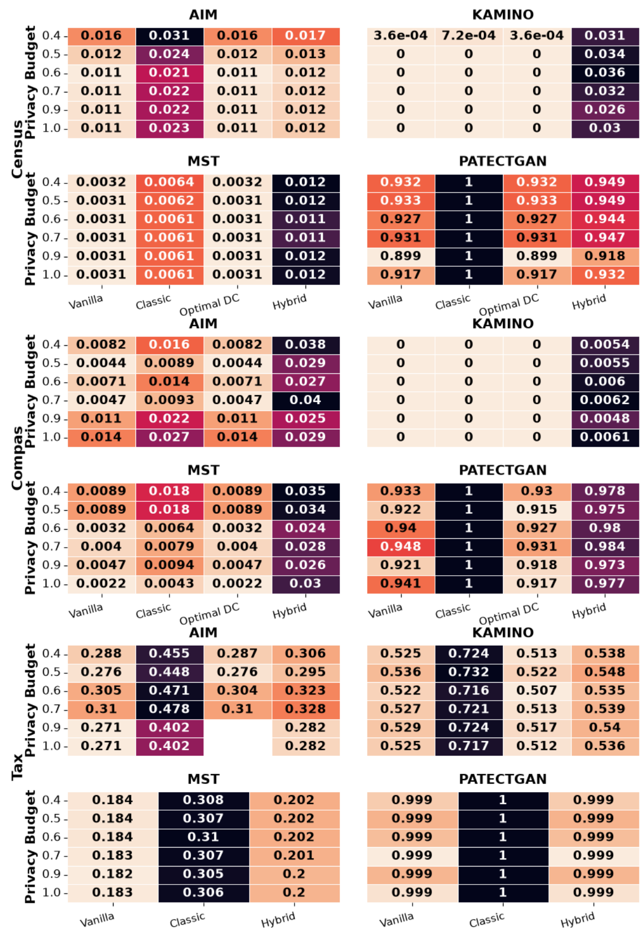


Figure 2: Deletion ratio as a function of the generation privacy budget ϵ. Optimal-DC-Baseline was not included in the heatmap when it reached the 12 hour timeout.

6.3 Repair Quality and Scalability Analysis

We evaluate the repair algorithms along three dimensions: repair cost (deletion ratio), statistical utility (marginal error), and runtime, as a function of the generator's privacy budget ϵ and the number of marginals, with runtime measured separately against synthetic dataset size on Census.

Deletion ratio. Figures 2 and 3 report the fraction of tuples removed during repair. As expected, Optimal-DC-Baseline achieves the lowest deletion ratio, while Vanilla-VC-Baseline closely matches the optimum. Since Hybrid additionally optimizes statistical fidelity, it performs slightly more deletions, whereas Classic-VC-Baseline often removes substantially more tuples because it ignores the topology of the conflict graph. For example, in Census AIM, Hybrid removes 1.3% of the data while Classic-VC-Baseline removes 2.4% of the data and Optimal-DC-Baseline and Vanilla-VC-Baseline remove 1.2%. As shown in Figure 2, the deletion ratio generally decreases as ϵ increases. Better synthetic data contains fewer denial constraint violations, reducing the amount of repair required. Figure 3 shows that increasing the number of marginals affects only Hybrid, since the remaining algorithms do not optimize statistical utility. In most settings, additional marginals result in a modest increase in the deletion ratio, reflecting the additional statistical constraints imposed during repair.

Marginal error. Figures 4 and 5 evaluate the statistical fidelity of the repaired datasets. Across all datasets and generation methods, Hybrid consistently achieves the lowest marginal error by explicitly optimizing the repair objective. As expected, marginal

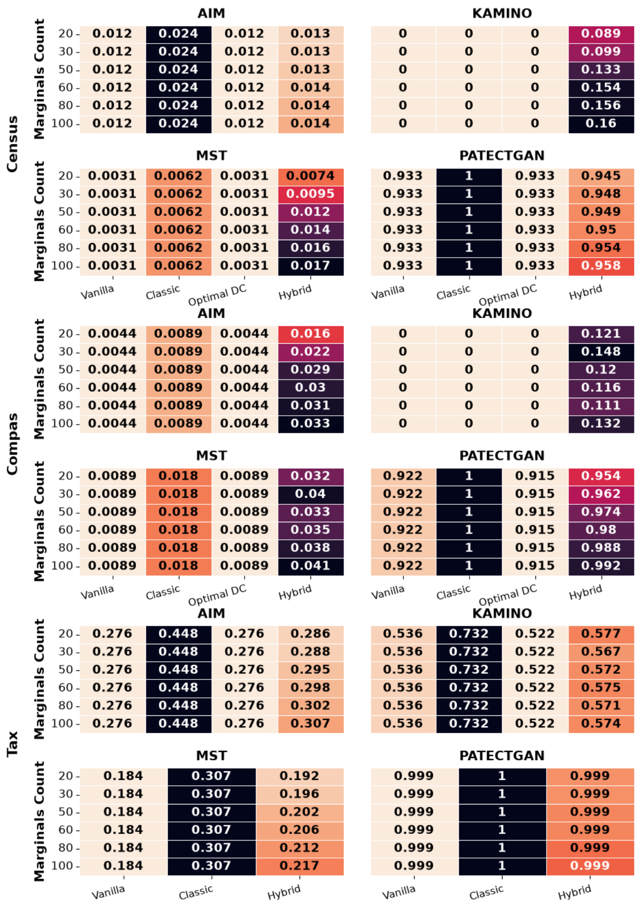


Figure 3: Deletion ratio as a function of the number of marginals. Optimal-DC-Baseline was not included in the heatmap when it reached the 12 hour timeout.

error generally decreases as ϵ increases because higher-quality synthetic datasets more closely approximate the private distribution. Conversely, increasing the number of optimized marginals slightly increases the overall error, since satisfying a larger collection of statistics becomes more challenging. Nevertheless, Hybrid consistently maintains substantially lower marginal error than all deletion-oriented repair methods. For example in Compas, PATE-CTGAN the marginal error of Hybrid is 0.488 while Optimal-DC-Baseline has an error of 49.7K, Vanilla-VC-Baseline has an error of 50K, the synthetic data has an error of 51.7K and Classic-VC-Baseline has an error of 54.5K.

Runtime. Figure 6 reports the runtime of the repair algorithms on the Census dataset for different synthetic data generation methods and dataset sizes. Runtime increases with the size of the repaired dataset, as larger datasets contain more tuples and typically induce more denial constraint violations. Among the heuristic methods, Hybrid incurs only a modest overhead over Vanilla-VC-Baseline as it additionally evaluates the statistical objective after each candidate deletion. Although Optimal-DC-Baseline solves an NP-hard optimization problem (Proposition 14), it remains competitive on several instances due to the effectiveness of modern ILP solvers. However, consistent with its worst-case complexity, several large

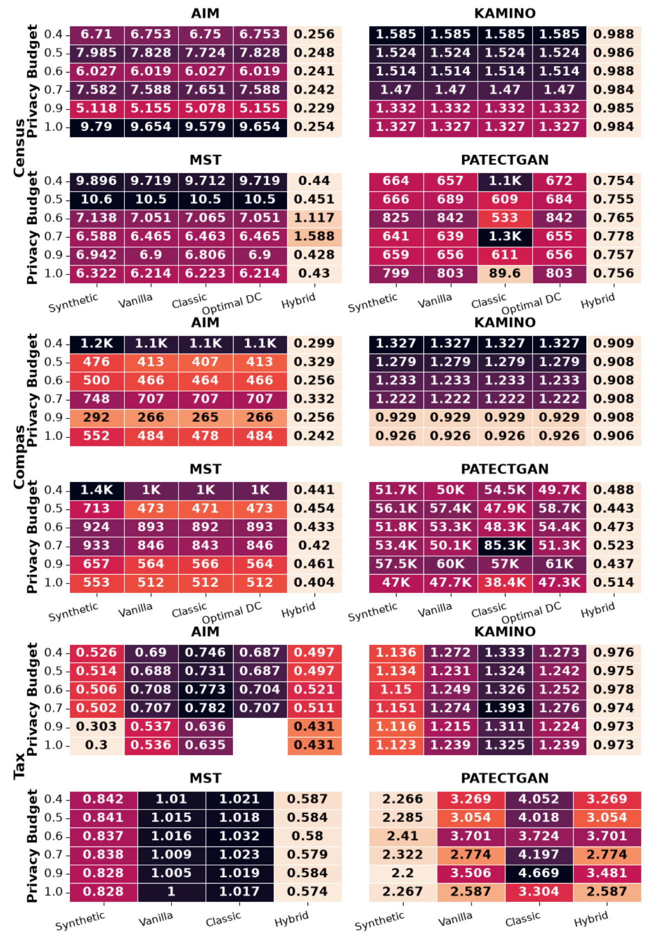


Figure 4: Marginal error as a function of the generation privacy budget ϵ. Optimal-DC-Baseline was not included in the heatmap when it reached the 12 hour timeout.

instances did not converge for over four hours and were therefore terminated. Overall, Hybrid provides substantially improved statistical utility while maintaining runtime comparable to existing scalable repair algorithms. For example, in PATE-CTGAN repairing a data with 1M tuples takes around 3 hours for Optimal-DC-Baseline around 1.5 hours for Classic-VC-Baseline and around 15 minutes for Vanilla-VC-Baseline and Hybrid .

6.4 Faithfulness to the Private Data

This section evaluates whether Hybrid improves the similarity of the repaired data to the original private dataset by comparing it *directly* to the private data, rather than to the noisy marginals supplied during repair as in the previous experiments. Specifically, we report the average two-way TVD and the marginal error with respect to the true private marginals, comparing Hybrid against the original synthetic dataset to illustrate the effect of the repair.

Two-way TVD. Figure 7 compares the average two-way TVD before and after repair across all datasets and generation methods. The repair process has only a modest effect on the total TVD of the synthetic data. Although Hybrid modifies the dataset to satisfy the

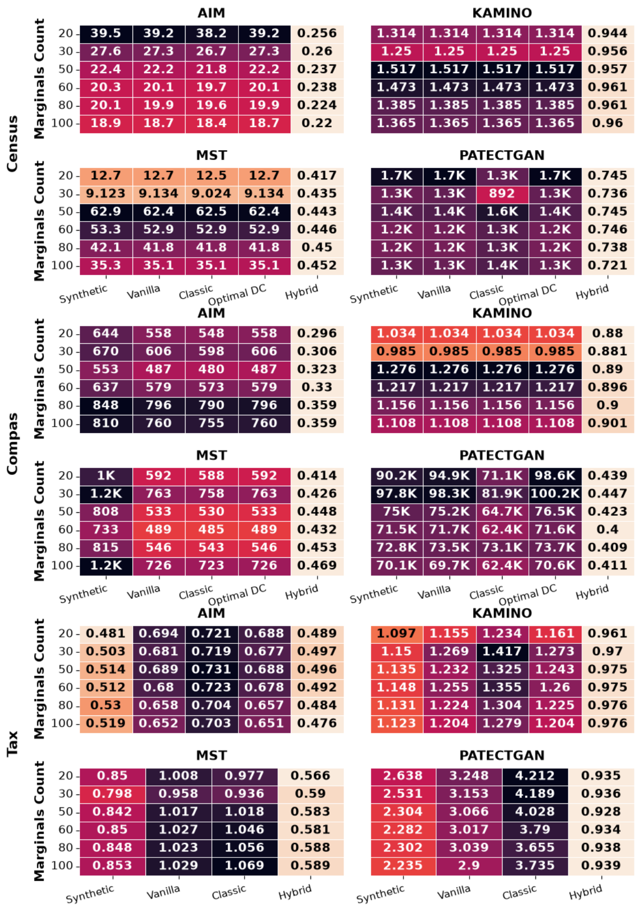


Figure 5: Marginal error as a function of the number of noisy marginals. Optimal-DC-Baseline was not included in the heatmap when it reached the 12 hour timeout.

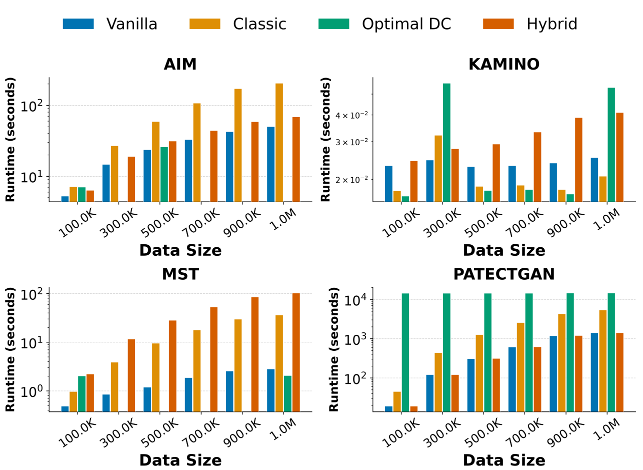


Figure 6: Runtime as a function of the synthetic data size for the Census dataset.

DCs and improve the selected marginals, the overall pairwise distribution remains largely unchanged. This indicates that the repair process preserves the global statistical properties of the synthetic

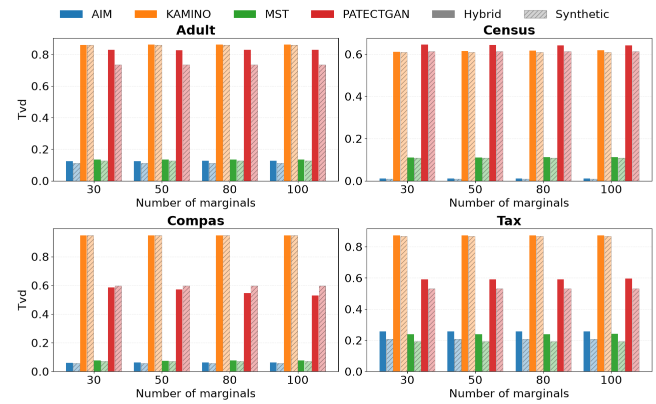


Figure 7: Two-way TVD as a function of number of marginals. The plots show that Hybrid usually matches the TVD of the synthetic data, and never drastically increases it.

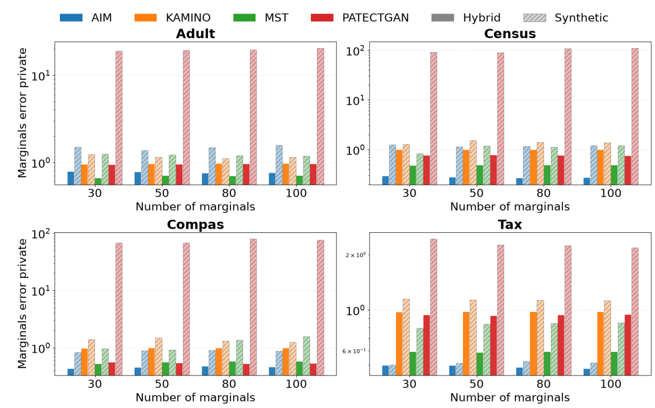


Figure 8: Marginal error with respect to the *marginals on the private data* as a function of the number of marginals. The graphs show that the marginal error compared with the private data marginals improves significantly from the starting synthetic data after the repair.

data while enforcing consistency. Figure 10 presents the TVD as a function of the privacy budget for marginal generation. The TVD remains mainly unaffected by the change of the budget. This makes sense as the set of marginals considered by the repair might be insignificant compared to the global TVD that looks at all marginals. For example, in the Tax data when choosing 100 marginals with budget $\rho = 1$ (Figure 7), the TVD after repairing the AIM data increases from 0.21 to 0.26, the TVD after repairing Kamino stays 0.87, the TVD after repairing MST increases from 0.19 to 0.24 and the TVD after repairing PATE-CTGAN increases from 0.53 to 0.59.

Marginal error with respect to the private data. Figure 8 reports the marginal error with respect to the true private marginals. Across nearly all datasets and generation methods, Hybrid consistently reduces the marginal error compared to the original synthetic data. This result demonstrates that optimizing noisy marginals successfully transfers to the underlying private distribution. Although the repair algorithm has access only to DP statistics, the repaired datasets capture the true statistics of the private data. Together with the TVD results, these experiments show that Hybrid substantially improves statistical fidelity while introducing only minor changes to the overall data distribution. Figure 11 presents the marginal error with respect to the private data as function of the privacy budget given to the marginals. As expected, the error decreases as the budget increases, meaning that as we increase the budget we get more faithful results to the private data, demonstrating the tradeoff between the privacy budget and the statistical fidelity. For example, in Adult when repairing with 80 marginals after repairing AIM the error decreased from 1.48 to 0.75, after repairing Kamino the error decreases from 1.11 to 0.96, when repairing MST the error decreases from 1.20 to 0.69 and when repairing PATE-CTGAN the error decreases from 19.6 to 0.96.

6.5 Hybrid Compared to the MILP

We have performed a comparison of Hybrid to the MILP approach presented in Section 4.1. We observe that the MILP achieves slightly lower marginal error at the prohibitive cost of lack of scalability and high deletion ratio. The MILP crossed a runtime of 4 hours on 138 of its 196 runs (70%), reaching a proven optimum in only 58 cases, and beyond 1,000 tuples it essentially never finishes: at sizes of 3,000 and over, every run times out and returns a degenerate solution that deletes the entire dataset. Even on the one size where both methods have substantial data (5,000 tuples), the MILP averages 11,600 seconds of runtime and a 0.7 deletion ratio, versus Hybrid's ~0.2 seconds and 0.39 deletion ratio. Across every size, Hybrid runs in well under a second, never times out, and keeps the deletion ratio stable around 0.4. On marginal error the two are essentially comparable (Hybrid ~0.73 on average versus the MILP ~0.6 on the small instances it can actually solve), so the MILP's slight marginal-error edge exists only where it is solvable at all, and it buys that edge by deleting far more tuples and running for hours. Hybrid delivers comparable marginal fidelity while retaining roughly 60% of the tuples in sub-second time.

6.6 Sensitivity to the Tradeoff Parameter α

The tradeoff parameter α balances data utility and statistical utility in the objective of Hybrid (Definition 11). Since Hybrid is the only repair algorithm that optimizes both objectives simultaneously, it is also the only method affected by α. Figure 9 shows that increasing α consistently reduces the deletion ratio while slightly increasing the marginals error, confirming that α provides the intended tradeoff between the two objectives. For example, in Census AIM, when $\alpha = 0$ the deletion ratio is around 8% of the data and the marginals error is 0.222, as α increases the deletion ratio decreases to 1.3% of the data and the marginals error increases to 0.244. Importantly, both metrics vary only modestly over a wide range of α values, indicating that Hybrid is robust to the precise choice of this parameter and does not require extensive tuning in practice. A detailed sensitivity analysis, including the behavior of the objective value and results for individual generation models, is deferred to Figure 9 .

7 RELATED WORK

We survey work on DP synthetic data generation and data repair. Our work is the first to complement and combine them via *post-hoc repair of DP synthetic data.*

Private synthetic data generation. A broad spectrum of techniques has been developed for generating DP synthetic data [24, 25, 28, 34, 35, 40, 41, 52, 54, 56–58]. These include methods that approximate the underlying distribution using low dimensional marginals [40–42, 58] as well as approaches based on GANs and other deep learning models [52, 54, 57]. Empirical studies [53] find that marginal-based systems such as MST [40], PrivBayes [58], and AIM [41] offer some of the strongest utility in the private setting. Kamino [21] extends this line of work by generating DP synthetic data that additionally satisfies (possibly soft) DCs [7], allowing users to incorporate integrity requirements directly into the synthesis pipeline. Despite these advances, generating high-quality DP synthetic data remains computationally expensive.

Data repair. Research on data repair spans a broad range of techniques aimed at restoring consistency with minimal modifications to the database [1, 3, 9, 18, 50]. Prior work has focused both on tuple deletions [1, 7, 23, 36, 37, 39, 45] and on value updates [3, 9, 11, 22, 27, 50], with the computational complexity of deletion-based repairs now relatively well-understood [37]. A recent line of work [36] has examined subset repairs under functional dependencies combined with representation constraints, which can be viewed as enforcing 1-way marginals. Our work considers the richer class of DCs [7] together with 2-way marginals, and re-frames the problem with a novel loss function.

Denial constraints and privacy. DCs are schema-level integrity constraints that capture universal validity rules over any instance of the schema (e.g., "two employees with the same role cannot have different salary grades"). Crucially, DCs *do not encode private information* about specific individuals: they are defined over the schema, not the data, and are typically derived from domain documentation or expert knowledge. Therefore, using DCs in our repair framework requires *no additional privacy budget.* This contrasts with the marginals, which are extracted from the private data and require DP treatment. Our framework's zero-budget DC assumption follows a standard convention in data-cleaning literature [9, 50] and is standard in prior work on private data repair [21].

8 CONCLUSION AND FUTURE WORK

We introduced *synthetic data repair under DP*, the first framework for post-hoc repair of DP synthetic data that enforces Denial Constraints while minimizing a weighted objective trading off deletions against relative marginal error. The problem is NP-hard even with the marginal terms alone or a single DC alone. We proposed an exact MILP and a scalable two-phase greedy heuristic; on four real-world datasets and four synthesizers, repair eliminates all DC violations and reduces the targeted marginal error, often by orders of magnitude, while largely preserving global fidelity and downstream ML utility. Future directions include richer statistical constraints and downstream fairness metrics (Section 1), settling the complexity of repair with constant-size DCs, alternative marginal selection and release mechanisms, and extending the deletion-based model to update-based repairs.

REFERENCES

[1] Foto N. Afrati and Phokion G. Kolaitis. 2009. Repair Checking in Inconsistent Databases: Algorithms and Complexity. In *ICDT*. 31–41.

[2] Leopoldo Bertossi. 2011. *Database Repairing and Consistent Query Answering*. Morgan & Claypool Publishers.

[3] Leopoldo E. Bertossi, Solmaz Kolahi, and Laks V. S. Lakshmanan. 2013. Data Cleaning and Query Answering with Matching Dependencies and Matching Functions. *Theory Comput. Syst.* 52, 3 (2013), 441–482.

[4] Philip Bohannon, Wenfei Fan, Floris Geerts, Xibei Jia, and Anastasios Kementsietsidis. 2007. Conditional functional dependencies for data cleaning. In *ICDE*.

[5] Mark Bun and Thomas Steinke. 2016. Concentrated differential privacy: Simplifications, extensions, and lower bounds. In *Theory of Cryptography Conference*. Springer, 635–658.

[6] Mark Cesar and Ryan Rogers. 2021. Bounding, concentrating, and truncating: Unifying privacy loss composition for data analytics. In *Algorithmic Learning Theory*. PMLR, 421–457.

[7] Jan Chomicki and Jerzy Marcinkowski. 2005. Minimal-change integrity maintenance using tuple deletions. *Inf. Comput.* 197, 1-2 (2005), 90–121.

[8] Xu Chu, Ihab F Ilyas, and Paolo Papotti. 2013. Discovering denial constraints. *Proceedings of the VLDB Endowment* 6, 13 (2013), 1498–1509.

[9] Xu Chu, Ihab F. Ilyas, and Paolo Papotti. 2013. Holistic data cleaning: Putting violations into context. In *ICDE*. 458–469.

[10] Kenneth L Clarkson. 1983. A modification of the greedy algorithm for vertex cover. *Inform. Process. Lett.* 16, 1 (1983), 23–25.

[11] Michele Dallachiesa, Amr Ebaid, Ahmed Eldawy, Ahmed K. Elmagarmid, Ihab F. Ilyas, Mourad Ouzzani, and Nan Tang. 2013. NADEEF: a commodity data cleaning system. In *SIGMOD*. 541–552.

[12] Jinshuo Dong, David Durfee, and Ryan Rogers. 2020. Optimal differential privacy composition for exponential mechanisms. In *International Conference on Machine Learning*. PMLR, 2597–2606.

[13] David Durfee and Ryan Rogers. 2019. Practical Differentially Private Top-k Selection with Pay-what-you-get Composition. arXiv:1905.04273 [cs.CR] https://arxiv.org/abs/1905.04273

[14] David Durfee and Ryan M. Rogers. 2019. Practical Differentially Private Top-k Selection with Pay-what-you-get Composition. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*. 3527–3537. https://proceedings.neurips.cc/paper/2019/hash/b139e104214a08ae3f2ebcce149cdf6e-Abstract.html

[15] Cynthia Dwork. 2006. Differential Privacy. In *Automata, Languages and Programming*, Michele Bugliesi, Bart Preneel, Vladimiro Sassone, and Ingo Wegener (Eds.). Springer Berlin Heidelberg.

[16] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating noise to sensitivity in private data analysis. In *Theory of cryptography conference*. Springer, 265–284.

[17] Cynthia Dwork, Aaron Roth, et al. 2014. The algorithmic foundations of differential privacy. *Found. Trends Theor. Comput. Sci.* 9, 3-4 (2014), 211–407.

[18] Ronald Fagin, Benny Kimelfeld, and Phokion G. Kolaitis. 2015. Dichotomies in the Complexity of Preferred Repairs. In *PODS*. 3–15.

[19] Sarah Flood, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, and Michael Westberry. 2022. Integrated Public Use Microdata Series, Current Population Survey: Version 10.0 [dataset]. Minneapolis, MN: IPUMS, 2022. https://doi.org/10.18128/D030.V10.0

[20] Georgi Ganev, Bristena Oprisanu, and Emiliano De Cristofaro. 2022. Robin Hood and Matthew Effects: Differential Privacy Has Disparate Impact on Synthetic Data. In *ICML*, Vol. 162. 6944–6959.

[21] Chang Ge, Shubhankar Mohapatra, Xi He, and Ihab F. Ilyas. 2021. Kamino: Constraint-Aware Differentially Private Data Synthesis. *Proc. VLDB Endow.* 14, 10 (2021), 1886–1899.

[22] Floris Geerts, Giansalvatore Mecca, Paolo Papotti, and Donatello Santoro. 2013. The LLUNATIC Data-Cleaning Framework. *Proc. VLDB Endow.* 6, 9 (2013), 625–636.

[23] Amir Gilad, Daniel Deutch, and Sudeepa Roy. 2020. On Multiple Semantics for Declarative Database Repairs. In *SIGMOD*. 817–831.

[24] Moritz Hardt and Guy N. Rothblum. 2010. A Multiplicative Weights Mechanism for Privacy-Preserving Data Analysis. *2010 IEEE 51st Annual Symposium on Foundations of Computer Science* (2010), 61–70.

[25] Zhiqi Huang, Ryan McKenna, George Bissias, Gerome Miklau, Michael Hay, and Ashwin Machanavajjhala. 2019. PSynDB: Accurate and Accessible Private Data Generation. *Proc. VLDB Endow.* 12, 12 (2019), 1918–1921. https://doi.org/10.14778/3352063.3352099

[26] Richard M. Karp. 1972. Reducibility Among Combinatorial Problems. In *Complexity of Computer Computations*, Raymond E. Miller and James W. Thatcher (Eds.). Plenum Press, New York, 85–103.

[27] Solmaz Kolahi and Laks V. S. Lakshmanan. 2009. On Approximating Optimum Repairs for Functional Dependency Violations. In *Proceedings of the 12th International Conference on Database Theory* (St. Petersburg, Russia) *(ICDT '09)*. Association for Computing Machinery, New York, NY, USA, 53–62. https://doi.org/10.1145/1514894.1514901

[28] Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. 2023. TabDDPM: Modelling Tabular Data with Diffusion Models. In *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA (Proceedings of Machine Learning Research)*, Vol. 202. PMLR, 17564–17579. https://proceedings.mlr.press/v202/kotelnikov23a.html

[29] Ios Kotsogiannis, Yuchao Tao, Xi He, Maryam Fanaeepour, Ashwin Machanavajjhala, Michael Hay, and Gerome Miklau. 2019. Privatesql: a differentially private sql query engine. *Proceedings of the VLDB Endowment* 12, 11 (2019), 1371–1384.

[30] Nick Koudas, Avishek Saha, Divesh Srivastava, and Suresh Venkatasubramanian. 2009. Metric functional dependencies. In *ICDE*.

[31] Chao Li and Gerome Miklau. 2012. An Adaptive Mechanism for Accurate Query Answering under Differential Privacy. *Proc. VLDB Endow.* 5, 6 (2012), 514–525. https://doi.org/10.14778/2168651.2168653

[32] Haoran Li, Li Xiong, Lifan Zhang, and Xiaoqian Jiang. 2014. DPSynthesizer: Differentially Private Data Synthesizer for Privacy Preserving Data Sharing. *Proc. VLDB Endow.* 7, 13 (2014), 1677–1680. https://doi.org/10.14778/2733004.2733059

[33] M. Lichman. 2013. UCI machine learning repository. https://archive.ics.uci.edu/ml/datasets/adult.

[34] Terrance Liu, Jingwu Tang, Giuseppe Vietri, and Steven Wu. 2023. Generating Private Synthetic Data with Genetic Algorithms. In *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA (Proceedings of Machine Learning Research)*, Vol. 202. PMLR, 22009–22027. https://proceedings.mlr.press/v202/liu23ag.html

[35] Terrance Liu, Giuseppe Vietri, and Steven Wu. 2021. Iterative Methods for Private Synthetic Data: Unifying Framework and New Methods. In *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*. 690–702. https://proceedings.neurips.cc/paper/2021/hash/0678c572b0d5597d2d4a6b5bd135754c-Abstract.html

[36] Yuxi Liu, Fangzhu Shen, Kushagra Ghosh, Amir Gilad, Benny Kimelfeld, and Sudeepa Roy. 2024. The Cost of Representation by Subset Repairs. *Proc. VLDB Endow.* 18, 2 (2024), 475–487. https://www.vldb.org/pvldb/vol18/p475-liu.pdf

[37] Ester Livshits, Benny Kimelfeld, and Sudeepa Roy. 2020. Computing Optimal Repairs for Functional Dependencies. *ACM Trans. Database Syst.* 45, 1 (2020), 4:1–4:46. https://doi.org/10.1145/3360904

[38] Ester Livshits, Rina Kochirgan, Segev Tsur, Ihab F. Ilyas, Benny Kimelfeld, and Sudeepa Roy. 2021. Properties of Inconsistency Measures for Databases. In *SIGMOD*. 1182–1194.

[39] Andrei Lopatenko and Leopoldo E. Bertossi. 2007. Complexity of Consistent Query Answering in Databases Under Cardinality-Based and Incremental Repair Semantics. In *ICDT*. 179–193.

[40] Ryan McKenna, Gerome Miklau, and Daniel Sheldon. 2021. Winning the NIST Contest: A scalable and general approach to differentially private synthetic data. arXiv:2108.04978 [cs.CR] https://arxiv.org/abs/2108.04978

[41] Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. 2022. AIM: An Adaptive and Iterative Mechanism for Differentially Private Synthetic Data. *CoRR* abs/2201.12677 (2022). arXiv:2201.12677 https://arxiv.org/abs/2201.12677

[42] Ryan McKenna, Daniel Sheldon, and Gerome Miklau. 2019. Graphical-model based estimation and inference for differential privacy. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA (Proceedings of Machine Learning Research)*, Vol. 97. PMLR, 4435–4444.

[43] Frank McSherry and Kunal Talwar. 2007. Mechanism Design via Differential Privacy. In *Proceedings of the 48th Annual IEEE Symposium on Foundations of Computer Science (FOCS '07)*. IEEE Computer Society, USA, 94–103. https://doi.org/10.1109/FOCS.2007.41

[44] Frank D McSherry. 2009. Privacy integrated queries: an extensible platform for privacy-preserving data analysis. In *SIGMOD*. 19–30.

[45] Dongjing Miao, Zhipeng Cai, Jianzhong Li, Xiangyu Gao, and Xianmin Liu. 2020. The computation of optimal subset repairs. *Proc. VLDB Endow.* 13, 12 (jul 2020), 2061–2074.

[46] Ilya Mironov. 2017. Rényi differential privacy. In *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*. IEEE, 263–275.

[47] Shubhankar Mohapatra, Amir Gilad, Xi He, and Benny Kimelfeld. 2025. Computing Inconsistency Measures Under Differential Privacy. *Proc. ACM Manag. Data* 3, 3 (2025), 140:1–140:27. https://doi.org/10.1145/3725397

[48] ProPublica. [n.d.]. Compas recidivism risk score data and analysis. https://www.propublica.org/datastore/dataset/compas-recidivism-risk-score-data-and-analysis

[49] Gang Qiao, Weijie J Su, and Li Zhang. 2021. Oneshot Differentially Private Top-k Selection. *arXiv preprint arXiv:2105.08233* (2021).

[50] Theodoros Rekatsinas, Xu Chu, Ihab F. Ilyas, and Christopher Ré. 2017. HoloClean: Holistic Data Repairs with Probabilistic Inference. *PVLDB* 10, 11 (2017), 1190–1201.

[51] Ryan Rogers and Thomas Steinke. 2021. A Better Privacy Analysis of the Exponential Mechanism. DifferentialPrivacy.org. https://differentialprivacy.org/exponential-mechanism-bounded-range/.

[52] Lucas Rosenblatt, Xiaoyan Liu, Samira Pouyanfar, Eduardo de Leon, Anuj Desai, and Joshua Allen. 2020. Differentially private synthetic data: Applied evaluations and enhancements. *arXiv preprint arXiv:2011.05537* (2020).
[53] Yuchao Tao, Ryan McKenna, Michael Hay, Ashwin Machanavajjhala, and Gerome Miklau. 2021. Benchmarking Differentially Private Synthetic Data Generation Algorithms. *CoRR* abs/2112.09238 (2021).
[54] Reihaneh Torkzadehmahani, Peter Kairouz, and Benedict Paten. 2019. DP-CGAN: Differentially Private Synthetic Data and Label Generation. In *CVPR*. 98–104.
[55] Vijay V Vazirani. 1997. Approximation algorithms. *Georgia Inst. Tech* (1997).
[56] Giuseppe Vietri, Cédric Archambeau, Sergül Aydöre, William Brown, Michael Kearns, Aaron Roth, Amaresh Ankit Siva, Shuai Tang, and Zhiwei Steven Wu. 2022. Private Synthetic Data for Multitask Learning and Marginal Queries. In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022.* http://papers.nips.cc/paper_files/paper/2022/hash/7428310c0f97f1c6bb2ef1be99c1ec2a-Abstract-Conference.html
[57] Jinsung Yoon, James Jordon, and Mihaela van der Schaar. 2019. PATE-GAN: Generating Synthetic Data with Differential Privacy Guarantees. In *International Conference on Learning Representations*. https://openreview.net/forum?id=S1zk9iRqF7
[58] Jun Zhang, Graham Cormode, Cecilia M Procopiuc, Divesh Srivastava, and Xiaokui Xiao. 2017. Privbayes: Private data release via bayesian networks. *ACM Transactions on Database Systems (TODS)* 42, 4 (2017), 1–41.

A HARDNESS PROOFS

A.1 Proof of Item (1) of Proposition 14

Proof. We reduce from the Max-Cut decision problem, which is NP-complete [26]. Given an undirected graph $G = (V, E)$ and an integer k with $1 \le k \le |E|$, Max-Cut asks whether there exists $S \subseteq V$ such that $\mathrm{cut}(S) = |\{(u, v) \in E : |S \cap \{u, v\}| = 1\}| \ge k$. Throughout, per Definition 3, the marginal of a non-empty repaired dataset D_r is normalized by $|D_r|$, and the marginals of the empty repair are 0 by convention, so each of its relative errors equals 1 and $\mathcal{L}(\emptyset) = 1$.

Construction. Let $n = |V|$ and $N = 4|E|\,n$. We create a relation D_s with schema $\mathcal{A}ttr = \{A_e : e \in E\} \cup \{X, B\}$, where each A_e and X are binary and $Dom(B) = \{0, 1, \dots, N\}$, containing: (i) a *vertex tuple* t_v for each $v \in V$, with $t_v[A_e] = 1$ iff $v \in e$, $t_v[X] = 1$, and $t_v[B] = 0$; and (ii) N *ballast tuples* $u_1, \dots, u_N$, with $u_i[A_e] = 0$ for all $e \in E$, $u_i[X] = 1$, and $u_i[B] = i$ (the attribute B only serves to make the ballast tuples pairwise distinct; it is not monitored). Thus $|D_s| = N + n$. We set $\Sigma = \emptyset$ (no DCs), $\alpha = 0$, and for each edge $e \in E$ define a marginal monitoring $(A_e{=}1, X{=}1)$ with private estimate $\tilde{m}_{D_p}(A_e{=}1, X{=}1) = 1/(N + n)$; thus $|\mathcal{M}| = |E|$. Since every private estimate is positive, the relative marginal errors in $\mathcal{L}$ are well-defined. The decision threshold for $\mathcal{L}(D_r)$ is $\lambda = (|E| - k + \frac{1}{2})/|E|$. The instance has $N + n = O(|E|\,n)$ tuples, $|E| + 2$ attributes, and $|E|$ marginals, so the reduction is polynomial.

The objective as a function of the repair. Consider first any non-empty repair $D_r \subseteq D_s$ retaining the vertex set $S = \{v : t_v \in D_r\}$ (with $s = |S|$) and $b \in [0, N]$ ballast tuples, so $|D_r| = b + s \ge 1$. For each edge e, only vertex tuples satisfy $A_e = 1$, hence the retained count for the monitored pair $(A_e{=}1, X{=}1)$ is $\kappa_e = |S \cap e| \in \{0, 1, 2\}$ and $m_{D_r}(A_e{=}1, X{=}1) = \kappa_e/(b + s)$. Writing $f = (N + n)/(b + s)$ and noting $b + s \le N + n$, we have $f \ge 1$ and, with $\alpha = 0$,

$$\mathcal{L}(D_r) \;=\; \frac{1}{|E|} \sum_{e \in E} \frac{\left|\kappa_e/(b+s) - 1/(N+n)\right|}{1/(N+n)} \;=\; \frac{1}{|E|} \sum_{e \in E} \left|\kappa_e f - 1\right|. \tag{5}$$

Since $f \ge 1$, each term equals 1 if $\kappa_e = 0$, equals $f - 1 \ge 0$ if $\kappa_e = 1$, and equals $2f - 1 \ge 1$ if $\kappa_e = 2$.

Lower bound (holds for every repair). From the case analysis above, every edge with $\kappa_e \in \{0, 2\}$ contributes at least 1 to the sum in (5), and edges with $\kappa_e = 1$ are exactly the edges cut by S. For the empty repair, every relative error equals 1 by convention, matching the $\kappa_e = 0$ case with $S = \emptyset$ and $\mathrm{cut}(\emptyset) = 0$. Hence, for *every* repair $D_r \subseteq D_s$,

$$|E| \cdot \mathcal{L}(D_r) \;\ge\; |E| - \mathrm{cut}(S). \tag{6}$$

($\Rightarrow$) Suppose $\mathrm{cut}(S) \ge k$ for some $S \subseteq V$, and take $D_r = \{t_v : v \in S\} \cup \{u_1, \dots, u_N\}$, i.e., retain S and *all* ballast tuples ($b = N$; note $D_r \ne \emptyset$ since $N \ge 1$). Then $f = (N + n)/(N + s)$, so $0 \le f - 1 = (n - s)/(N + s) \le n/N = 1/(4|E|)$. Splitting the sum in (5) by the value of κ_e and using $2f - 1 = 2(f - 1) + 1$ to split each doubly-covered edge's term, so that the trailing $+1$ (one per edge with $\kappa_e{=}2$) joins the $\kappa_e{=}0$ edges in the constant, while $\mathrm{cut}(S) = \#\{\kappa_e{=}1\}$ gives $\#\{\kappa_e{=}0\} + \#\{\kappa_e{=}2\} = |E| - \mathrm{cut}(S)$, we obtain

$$\begin{aligned}
|E| \cdot \mathcal{L}(D_r) &= \#\{\kappa_e{=}0\} + \#\{\kappa_e{=}1\}\,(f-1) + \#\{\kappa_e{=}2\}\,(2f-1) \\
&= \underbrace{\big(\#\{\kappa_e{=}0\} + \#\{\kappa_e{=}2\}\big)}_{=|E|-\mathrm{cut}(S)} + \big(\#\{\kappa_e{=}1\} + 2\,\#\{\kappa_e{=}2\}\big)(f-1) \\
&\le \big(|E| - \mathrm{cut}(S)\big) + 2|E| \cdot \frac{1}{4|E|} \;\le\; |E| - k + \tfrac{1}{2},
\end{aligned}$$

hence $\mathcal{L}(D_r) \le \lambda$.

($\Leftarrow$) Suppose $\mathcal{L}(D_r^*) \le \lambda$ for some $D_r^* \subseteq D_s$, and let $S^* = \{v : t_v \in D_r^*\}$. By the lower bound (6), $|E| - \mathrm{cut}(S^*) \le |E| \cdot \mathcal{L}(D_r^*) \le |E| \cdot \lambda = |E| - k + \frac{1}{2}$, so $\mathrm{cut}(S^*) \ge k - \frac{1}{2}$, and since $\mathrm{cut}(S^*)$ is an integer, $\mathrm{cut}(S^*) \ge k$.

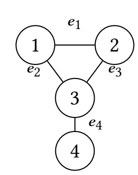
Since Max-Cut is NP-complete and the construction uses $\Sigma = \emptyset$, the decision version of Definition 13 is NP-hard even without DCs; the hardness arises entirely from the marginal terms of the objective. Membership in NP is immediate (a consistent repair D_r is a polynomial certificate and $\mathcal{L}(D_r)$ is computable in polynomial time), so the decision version is in fact NP-complete. □

Intuitively, the lower bound (6) holds for *every* repair, regardless of how many ballast tuples it retains; the ballast is needed only in the ($\Rightarrow$) direction, where retaining all of it pins the normalizing denominator $|D_r|$ close to $N + n$, so that edges with exactly one retained endpoint contribute a negligible relative error ($f - 1 \le 1/(4|E|)$). Without the ballast, the denominator $|D_r| = |S|$ varies with the chosen cut and the objective is no longer a monotone function of $\mathrm{cut}(S)$.

X	B	A_{e_1}	A_{e_2}	A_{e_3}	A_{e_4}
1	0	1	1	0	0
1	0	1	0	1	0
1	0	0	1	1	1
1	0	0	0	0	1
1	i	0	0	0	0

Table 5: Relation for the reduction of Item (1) of Proposition 14 from Max-Cut (see Example 23). The first four rows are the vertex tuples ($A_e = 1$ iff the vertex is an endpoint of edge e); the last row stands for the $N = 64$ ballast tuples u_i, $i \in [1, N]$.

Example 23. To illustrate Item (1) of Proposition 14, consider the graph G with vertices $\{v_1, v_2, v_3, v_4\}$ and edges $e_1 = (v_1, v_2)$, $e_2 = (v_1, v_3)$, $e_3 = (v_2, v_3)$, $e_4 = (v_3, v_4)$.



Here $n = 4$ and $|E| = 4$, so the construction sets $N = 4|E|n = 64$, $\alpha = 0$, and targets $\tilde{m}_{D_p}(A_e{=}1, X{=}1) = 1/68$, producing the relation in Table 5 (68 tuples in total). The maximum cut of G has value 3, achieved e.g. by $S = \{v_1, v_2, v_4\}$, which cuts e_2, e_3, e_4. Retaining S together with all 64 ballast tuples gives $|D_r| = 67$ and $f = 68/67$, so the four monitored relative errors are $|2f - 1| = 69/67$ (for e_1, with both endpoints retained) and $f - 1 = 1/67$ (for each of e_2, e_3, e_4), yielding $\mathcal{L}(D_r) = \frac{1}{4}\big(\frac{69}{67} + \frac{3}{67}\big) = \frac{18}{67} \approx 0.269$. For $k = 3$ the threshold is $\lambda = (4 - 3 + \frac{1}{2})/4 = 0.375 \ge \mathcal{L}(D_r)$, certifying a cut of size at least 3; for $k = 4$ we have $\lambda = 0.125$, and indeed an exhaustive check confirms that no repair attains $\mathcal{L} \leq 0.125$, matching the fact that G has no cut of size 4. In contrast, dropping the ballast breaks the encoding. Retaining only $S = \{v_1, v_3\}$ (also a maximum cut, cutting e_1, e_3, e_4 while e_2 is doubly covered) gives $|D_r| = 2$, so $f = (N+n)/|D_r| = 68/2 = 34$ and the four monitored relative errors are $|2f-1| = 67$ for e_2 ($\kappa = 2$) and $|f-1| = 33$ for each of e_1, e_3, e_4 ($\kappa = 1$), yielding $\mathcal{L}(D_r) = \frac{1}{4}(33+67+33+33) = 41.5$. This is *worse* than retaining all four vertex tuples without ballast ($|D_r| = 4$, $f = 17$, every edge doubly covered, $\mathcal{L} = \frac{1}{4}(4 \cdot 33) = 33$), and worse still than the empty repair ($\mathcal{L}(\emptyset) = 1$). Hence, without ballast, minimizing $\mathcal{L}$ would return the empty repair—encoding a cut of 0—rather than the maximum cut of value 3, so the objective no longer tracks the cut. (The value $\mathcal{L} = 1$ for retaining all four vertex tuples is attained only when the 64 ballast tuples are *also* kept, giving $f = 1$; that is the with-ballast regime used above, not a no-ballast repair.)

A.2 Proof of Item (2) of Proposition 14

Proof. We reduce from the Independent Set decision problem, which is NP-complete [26]: given an undirected graph $G = (V, E)$ and an integer K with $1 \leq K \leq |V|$, decide whether G has an independent set of size at least K. Let $n = |V|$ and let $\bar{E} = \{\{u, v\} : u, v \in V,\ u \neq v,\ \{u, v\} \notin E\}$ denote the set of non-edges of G.

Construction. We create a relation D_s with schema $\mathcal{A}ttr = \{P, Q, A_0\} \cup \{A_f : f \in \bar{E}\}$ containing: (i) a *vertex tuple* t_v for each $v \in V$, with $t_v[P] = 1$, $t_v[Q] = 1$, $t_v[A_0] = v$, and, for each non-edge $f \in \bar{E}$, $t_v[A_f] = f$ if $v \in f$ and $t_v[A_f] = \perp_v$ otherwise, where $\perp_v$ is a value unique to v and distinct from all non-edge labels; and (ii) $N = n$ *ballast tuples* $u_1, \ldots, u_N$ with $u_i[P] = 0$, $u_i[Q] = 1$, and a fresh constant on all remaining attributes. Thus $|D_s| = N + n = 2n$. We set $\alpha = 1/2$ and define the single DC

$$\sigma : \forall t_1, t_2\ \neg\Big(t_1[P] = 1 \wedge t_2[P] = 1 \wedge t_1[A_0] \neq t_2[A_0] \wedge \bigwedge_{f \in \bar{E}} t_1[A_f] \neq t_2[A_f]\Big),$$

which uses only predicates of the forms $t_i[A_j] \circ v$ and $t_1[A_j] \circ t_2[A_j]$ permitted by our DC class and has $|\bar{E}|+3 = O(n^2)$ predicates. Finally, we define a single marginal monitoring the value pair (P=1, Q=1) with private estimate $\tilde{m}_{D_p}(P{=}1, Q{=}1) = K/(N+K) > 0$, and the decision threshold $\lambda = \alpha\,(n-K)/(N+n)$. The construction is polynomial in $|G|$.

The conflict graph of σ is exactly G. Ballast tuples never participate in a violation, as they fail the predicate $t[P] = 1$. Consider two distinct vertex tuples t_u, t_v. They always disagree on A_0 (its values are unique per vertex), and they agree on A_f iff $t_u[A_f] = t_v[A_f] = f$, which holds iff $f = \{u, v\} \in \bar{E}$. Hence t_u, t_v jointly violate σ iff they disagree on *every* attribute in the conjunction, i.e., iff $\{u, v\} \notin \bar{E}$, i.e., iff $\{u, v\} \in E$. (No tuple conflicts with itself, since every tuple agrees with itself on A_0.) Consequently, a repair $D_r \subseteq D_s$ is consistent iff its retained vertex set $S = \{v : t_v \in D_r\}$ is an independent set of G, while the number $b \in [0, N]$ of retained ballast tuples is unconstrained.

The objective. All tuples satisfy $Q = 1$ and exactly the vertex tuples satisfy $P = 1$, so a non-empty repair D_r keeping S (with $s = |S|$) and b ballast tuples has $m_{D_r}(P{=}1, Q{=}1) = s/(b+s)$ and

$$\mathcal{L}(D_r) = \alpha \cdot \frac{(N+n)-(b+s)}{N+n} + (1-\alpha) \cdot \frac{\left|s/(b+s) - \frac{K}{N+K}\right|}{K/(N+K)}.$$

($\Rightarrow$) Suppose G has an independent set of size at least K. Retain exactly K of its vertices (any subset of an independent set is independent) together with all N ballast tuples. Then $s/(b+s) = K/(N+K)$, so the marginal term vanishes, and $\mathcal{L}(D_r) = \alpha\,(n-K)/(N+n) = \lambda$.

($\Leftarrow$) Suppose every independent set of G has size at most $K-1$, and consider any consistent non-empty repair, with $s \leq K-1$ retained vertex tuples and $b \leq N$ retained ballast tuples. Then $b + s \leq N + K - 1$, so the deletion term alone is at least $\alpha\big((N+n) - (N+K-1)\big)/(N+n) = \lambda + \alpha/(N+n) > \lambda$, and since the marginal term is non-negative, $\mathcal{L}(D_r) > \lambda$. The empty repair has $\mathcal{L}(\emptyset) = 1 > \lambda$ as well. Hence no consistent repair attains $\mathcal{L} \leq \lambda$.

Since Independent Set is NP-complete and the construction is polynomial, the decision version of Definition 13 is NP-hard (indeed NP-complete, with membership as in the proof of Item (1)) already for $|\Sigma| = |\mathcal{M}| = 1$. The argument applies verbatim for every fixed $\alpha \in (0, 1]$. □

A.3 Proof of Item (3) of Proposition 14

Proof. We reuse the vertex tuples and the single DC σ from the proof of Item (2), omitting the ballast tuples, and set $\mathcal{M} = \emptyset$, $\alpha = 1$, and threshold $\lambda = (n-K)/n$. By the convention of Definition 11, the marginal term vanishes when $\mathcal{M} = \emptyset$, so $\mathcal{L}(D_r) = (n - |D_r|)/n$ and $\mathcal{L}(D_r) \leq \lambda$ iff $|D_r| \geq K$. As shown in the proof of Item (2), a repair is consistent iff its retained vertex set is an independent set of G; hence a consistent repair of size at least K exists iff G has an independent set of size at least K (any subset of an independent set is independent). The empty repair has $\mathcal{L}(\emptyset) = \alpha = 1 > \lambda$ since $K \geq 1$. NP-hardness (and NP-completeness) follows as before. Note that α is part of the input and the instance above fixes $\alpha = 1$; any fixed $\alpha > 0$ works by scaling λ accordingly, whereas for $\alpha = 0$ and $\mathcal{M} = \emptyset$ the objective is identically 0 and the problem is trivial. □

B PROOF OF PROPOSITION 15

Proof. Let D_s be a (non-empty) synthetic database with $n = |D_s| \geq 1$, Σ a set of DCs, $\mathcal{M}$ a set of marginals, and $\alpha \in [0, 1]$. The MILP variables are $x_l \in \{0, 1\}$ for $t_l \in D_s$, $d_m \geq 0$ for $m \in \mathcal{M}$, and $z_{ml} \geq 0$ for $m \in \mathcal{M}$, $l \in [1, n]$. For a feasible point (x, d, z) we write $D_r(x) = \{t_l \mid x_l = 1\}$ and, for $m = (A_i{=}a_1, A_j{=}a_2)$, $C_m(x) = \sum_{l:\, t_l[A_i]=a_1,\, t_l[A_j]=a_2} x_l$. We take as a *standing precondition on the MILP input* that each private estimate satisfies $\tilde{m}_{D_p}(a_1, a_2) \in (0, 1]$. This is an assumption on the targets alone, *independent of how they were produced*: positivity is required by Definition 10, and the upper bound $\tilde{m}_{D_p}(a_1, a_2) \leq 1$ holds for any target after the source-independent preprocessing that clips each estimate to $[0, 1]$ (the remark following this proof shows this clipping never increases the target's error, and gives a variant of the program that dispenses with it entirely and admits arbitrary $\tilde{m}_{D_p}(a_1, a_2) > 0$). In particular the DP mechanism of Algorithm 6 already returns estimates in $[0, 1]$, but no step of this proof relies on that mechanism. Recall also that, by the convention of Definition 3, the marginals of the empty repair are 0, so each of its relative errors equals 1 and $\mathcal{L}(\emptyset) = 1$.

Step 1: exactness of the linearization. At every feasible point, $z_{ml} = d_m \cdot x_l$: if $x_l = 1$, then $z_{ml} \leq d_m$ and $z_{ml} \geq d_m - (1 - x_l) = d_m$ give $z_{ml} = d_m$; if $x_l = 0$, then $z_{ml} \leq x_l = 0$ and $z_{ml} \geq 0$ give $z_{ml} = 0$. Consequently $\sum_l z_{ml} = d_m \sum_l x_l$ at every feasible point.

(When $x_l = 0$, the constraint $z_{ml} \geq d_m - 1$ additionally forces $d_m \leq 1$; by Step 3 this never excludes an intended solution, since the repaired marginal lies in $[0,1]$ and, by the input precondition, so does each private estimate $\tilde{m}_{D_p}(a_1, a_2)$, so every required distance $|m_{D_r}(A_i{=}a_1, A_j{=}a_2) - \tilde{m}_{D_p}(a_1, a_2)|$ is at most 1.)

Step 2: every feasible point over-approximates $\mathcal{L}$. Let (x, d, z) be feasible and $D_r = D_r(x)$. First, $D_r \models \Sigma$: if a pair $(t_i, t_j) \in D_r$ violated some $\sigma \in \Sigma$, then $x_i = x_j = 1$ would give $x_i + x_j = 2 > 1$, contradicting feasibility. Second, we claim $d_m / \tilde{m}_{D_p}(a_1, a_2) \geq E_m(m_{D_r}(A_i{=}a_1, A_j{=}a_2))$ for every m. If $\sum_l x_l \geq 1$, by Step 1 the two distance constraints read $d_m \sum_l x_l \geq \pm\big(C_m(x) - \tilde{m}_{D_p}(a_1, a_2) \sum_l x_l\big)$; dividing by $\sum_l x_l \geq 1 > 0$ yields $d_m \geq |\, C_m(x)/\sum_l x_l - \tilde{m}_{D_p}(a_1, a_2)\,| = |m_{D_r}(A_i{=}a_1, A_j{=}a_2) - \tilde{m}_{D_p}(a_1, a_2)|$, proving the claim. If instead $\sum_l x_l = 0$, i.e., $D_r = \emptyset$, the safeguard constraint gives $d_m \geq \tilde{m}_{D_p}(a_1, a_2)$, so $d_m / \tilde{m}_{D_p}(a_1, a_2) \geq 1 = E_m(m_\emptyset(A_i{=}a_1, A_j{=}a_2))$, proving the claim as well. Since also $(n - \sum_l x_l)/n = (n - |D_r|)/n$, the MILP objective at (x, d, z) is at least $\mathcal{L}(D_r)$ in both cases.

Step 3: every repair is realized with objective exactly $\mathcal{L}$. Conversely, let D_r be any DC-consistent subset of D_s, possibly empty. Set $x_l = \mathbb{1}[t_l \in D_r]$, $d_m = |m_{D_r}(A_i{=}a_1, A_j{=}a_2) - \tilde{m}_{D_p}(a_1, a_2)|$, and $z_{ml} = d_m x_l$ for each $m \in \mathcal{M}$, $l \in [1, n]$ (for $D_r = \emptyset$ this reads $x = 0$, $d_m = \tilde{m}_{D_p}(a_1, a_2)$, $z = 0$). Then $d_m \in [0, 1]$, since the repaired marginal lies in $[0, 1]$ and, by the input precondition, so does the private estimate $\tilde{m}_{D_p}(a_1, a_2)$. All constraints hold: the DC constraints because D_r is consistent; the two distance constraints with equality, since $\sum_l z_{ml} = d_m \sum_l x_l = |C_m(x) - \tilde{m}_{D_p}(a_1, a_2) \sum_l x_l|$ (both sides are 0 when $D_r = \emptyset$); the safeguard constraints, with equality when $D_r = \emptyset$ and with non-positive right-hand side otherwise; and the linearization constraints directly (for $x_l = 0$, $z_{ml} = 0 \geq d_m - 1$ uses $d_m \leq 1$). The MILP objective at this point equals $\mathcal{L}(D_r)$ exactly (in particular, $\alpha + (1 - \alpha) = 1 = \mathcal{L}(\emptyset)$ when $D_r = \emptyset$).

Step 4: conclusion. Step 3 applied to $D_r = \emptyset$ shows that the MILP is always feasible. Let (x^*, d^*, z^*) be an optimal solution with objective value OPT, and let $D_r^* = D_r(x^*)$. For every DC-consistent $D_r \subseteq D_s$,

$$\mathcal{L}(D_r^*) \overset{\text{Step 2}}{\leq} \text{OPT} \overset{\text{Step 3}}{\leq} \mathcal{L}(D_r).$$

Hence D_r^* minimizes $\mathcal{L}$ over all DC-consistent subsets of D_s, i.e., it is an optimal synthetic data repair under Definition 11. Moreover, taking $D_r = D_r^*$ shows $\text{OPT} = \mathcal{L}(D_r^*)$. Note that the argument covers all $\alpha \in [0, 1]$: no step relies on d_m having a strictly positive coefficient in the objective. □

Remark (agnostic to the target source). The proof of Proposition 15 uses only one property of the private estimates, namely the input precondition $\tilde{m}_{D_p}(a_1, a_2) \in (0, 1]$; it never refers to Algorithm 6 or to any particular releasing mechanism. Correctness is therefore independent of the target source, consistent with the framing in Section 5: the targets may be analyst-specified, reused from statistics the synthesizer already released, or produced by the DP procedure of Algorithm 6. Two observations make the precondition harmless for an arbitrary source.

(i) Clipping to $[0, 1]$ is lossless. Any target can be forced to satisfy the precondition by the deterministic preprocessing $\tilde{m}_{D_p}(a_1, a_2) \mapsto \min\{1, \tilde{m}_{D_p}(a_1, a_2)\}$ (values are already positive). Since every attainable marginal of D_p lies in $[0, 1]$, this map is the Euclidean projection onto the interval that contains the ground-truth value $m_{D_p}(A_i{=}a_1, A_j{=}a_2)$. Projection onto a convex set is non-expansive, so $|\min\{1, \tilde{m}_{D_p}(a_1, a_2)\} - m_{D_p}(A_i{=}a_1, A_j{=}a_2)| \leq |\tilde{m}_{D_p}(a_1, a_2) - m_{D_p}(A_i{=}a_1, A_j{=}a_2)|$: clipping never increases the target's error with respect to the true marginal, and therefore only weakens the noise a downstream repair must accommodate. Because clipping depends on neither the estimate's provenance nor its noise model, it is a generic, source-independent front-end; Algorithm 6 already applies it internally (Line 14), but an external source is handled identically.

(ii) Targets outside $[0, 1]$ need no clipping. If one prefers to consume a raw estimate $\tilde{m}_{D_p}(a_1, a_2) > 1$ verbatim, the program of Equation (4) extends without any change to its optimum. Fix, for each m, the a-priori bound $U_m := \max\{1, \tilde{m}_{D_p}(a_1, a_2)\}$, and replace the linearization constraints $z_{ml} \leq x_l$ and $z_{ml} \geq d_m - (1 - x_l)$ by

$$z_{ml} \leq U_m x_l, \qquad z_{ml} \geq d_m - U_m(1 - x_l),$$

keeping $z_{ml} \leq d_m$ and $z_{ml} \geq 0$. This is the exact McCormick envelope of $z_{ml} = d_m x_l$ over $d_m \in [0, U_m]$: for binary x_l it still forces $z_{ml} = d_m$ when $x_l = 1$ and $z_{ml} = 0$ (with $d_m \leq U_m$) when $x_l = 0$. The bound U_m is valid because the required distance never exceeds it: $|m_{D_r}(A_i{=}a_1, A_j{=}a_2) - \tilde{m}_{D_p}(a_1, a_2)| \leq \max\{\tilde{m}_{D_p}(a_1, a_2),\ 1 - 0\} \leq U_m$ for every non-empty repair, and the empty-repair distance equals $\tilde{m}_{D_p}(a_1, a_2) \leq U_m$. Hence the safeguard $d_m \geq \tilde{m}_{D_p}(a_1, a_2)(1 - \sum_l x_l)$ can no longer conflict with the linearization, so the MILP remains feasible (the empty repair is realized as in Step 3) and exact (Steps 1–3 go through verbatim with 1 replaced by U_m) for any $\tilde{m}_{D_p}(a_1, a_2) > 0$. When $\tilde{m}_{D_p}(a_1, a_2) \leq 1$ we have $U_m = 1$ and the formulation reduces to Equation (4).

C EFFICIENT INCREMENTAL LOSS CALCULATION

Algorithm 2: State Initialization

Input : Dataset D, Marginal set $\mathcal{M}$ with private estimates $\{\tilde{m}_{D_p}(a_i, a_j)\}$, parameters $\alpha \in [0, 1]$

Output : Counts c, Indicator map $\mathcal{I}$, current size n_{cur}, original size $n = |D|$

```
// Materialize counts for each marginal
for m ∈ M over (A_i=a_i, A_j=a_j) do
    c_m ← |{t ∈ D | t[A_i] = a_i ∧ t[A_j] = a_j}|;
// Pre-compute tuple-constraint indicator
for t ∈ D do
    for m ∈ M do
        I(t, m) ← 𝟙[t[A_i] = a_i ∧ t[A_j] = a_j];
n_cur ← |D|;
return (c, I, n_cur);
```

Incremental Maintenance Strategy. A naive recalculation of the loss function in a hypothetical removal of a tuple at each iteration necessitates a full dataset scan, incurring a prohibitive $O(n \cdot |\mathcal{M}|)$ computational cost that severely limits scalability. To mitigate this, we propose an incremental maintenance strategy that decouples the loss calculation from the raw data by materializing the satisfaction states of the constraints. Our strategy follows a three-phase lifecycle: initialization, calculation, and state update, as detailed in Algorithms 2 to 4.

Algorithm 3: Loss Calculation on Hypothetical Removal

Input : Set of k tuples $\{t_1, \ldots, t_k\}$, Counts c, Indicator map $\mathcal{I}$, current size n_{cur}, private estimates $\{\tilde{m}_{D_p}(a_i, a_j)\}$, parameters α, $|\mathcal{M}|$, n

Output: Loss $\mathcal{L}(D_r \setminus \{t_1, \ldots, t_k\})$

$n_{new} \leftarrow n_{cur} - k$;

$m_{new} \leftarrow \frac{c_m - \sum_{i=1}^{k} \mathcal{I}(t_i, m)}{n_{new}}$;

$E_m \leftarrow \frac{|m_{new} - \tilde{m}_{D_p}(a_i, a_j)|}{\tilde{m}_{D_p}(a_i, a_j)}$;

return $\alpha \cdot \frac{n - n_{new}}{n} + (1 - \alpha) \cdot \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} E_m$;

Algorithm 4: Incremental State Update

Input : Removed tuples $t_1^*, \ldots, t_k^*$, counts c, indicator map $\mathcal{I}$, current size n_{cur}

Output: Updated counts c, updated size n_{cur}

// Decrement counts for affected marginals

for $m \in \mathcal{M}$ **do**

 $c_m \leftarrow c_m - \sum_{i=1}^{k} \mathcal{I}(t_i^*, m)$

$n_{\text{cur}} \leftarrow n_{\text{cur}} - k$;

return $(c,\ n_{cur})$;

State Initialization (Algorithm 2). To avoid expensive data access during iterative refinement, we materialize the initial state of the dataset. We compute a frequency vector c, where each entry c_m stores the absolute count of tuples related to a marginal $m \in \mathcal{M}$ (i.e., $c_m = |\{t \in D \mid t[A_i] = a_i \wedge t[A_j] = a_j\}|$ for the marginal over $(A_i = a_i, A_j = a_j)$). In practice, this is efficiently computed via indexed `GROUP BY` operations. We also store the noisy private estimates $\tilde{m}_{D_p}(a_i, a_j)$ for each $m \in \mathcal{M}$ and the current dataset size $n_{\text{cur}} = |D_s|$. Furthermore, we pre-calculate an indicator matrix $\mathcal{I} : D \times \mathcal{M} \rightarrow \{0, 1\}$, defined as $\mathcal{I}(t, m) = \mathbb{1}[t[A_i] = a_i \wedge t[A_j] = a_j]$. By storing $\mathcal{I}$ as a sparse index, we ensure $O(1)$ lookup complexity during subsequent phases with minimal memory overhead.

Loss Calculation (Algorithm 3). As defined in Definition 11, the loss function reflects the weighted sum between the deletion ratio and the marginals error. At the hypothetical removal of tuples $t_1, \ldots, t_k$, the new size of the data would be $n_{new} = n_{cur} - k$, and the deletion ratio can be easily calculated by $\frac{n - n_{new}}{n}$. For marginal $m \in \mathcal{M}$ we can calculate the error as

$$\frac{\left|\frac{c_m - \sum_{i=1}^{k} \mathcal{I}(t_i, m)}{n_{new}} - \tilde{m}_{D_p}(a_i, a_j)\right|}{\tilde{m}_{D_p}(a_i, a_j)} \tag{7}$$

This is because $c_m - \mathcal{I}(t, m)$ represents the number of tuples in the data related to m after t was removed, thus $\dfrac{c_m - \mathcal{I}(t, m)}{n_{new}}$ represents the value of m in the data after the removal. This operation is repeated for all marginals and the results are averaged to get the average relative error. For each marginal we do an $O(1)$ operation thus the complexity of the entire calculation of $\mathcal{L} \setminus \{t\}$ is $O(|\mathcal{M}|)$.

State Update (Algorithm 4). Upon selecting tuples $t_1^*, \ldots, t_k^*$ for removal, we perform a localized update to the materialized state. For each $m \in \mathcal{M}$ we decrement $c_m \leftarrow c_m - \sum_{i=1}^{k} \mathcal{I}(t_i, m)$. The current dataset size is decremented: $n_{\text{cur}} \leftarrow n_{\text{cur}} - k$. Since $\tilde{m}_{D_p}(a_i, a_j)$ is fixed (a noisy private estimate, not recomputed during repair), all the information needed for the next iteration is already given by this update. No full dataset scan is needed.

Complexity Analysis. The initialization (Algorithm 2) is a one-time cost of $O(n \cdot |\mathcal{M}|)$. Crucially, for each subsequent iteration, the weight calculation (Algorithm 3) iterates over the marginals and computes a single arithmetic expression, running in $O(|\mathcal{M}|)$ time. Note that we consider k as a constant. The state update (Algorithm 4) decrements the relevant counts and the dataset size, also in $O(|\mathcal{M}|)$ time. Since $|\mathcal{M}|$ is typically small and independent of n, this incremental strategy to calculate the loss achieves near-constant time updates per tuple removal, ensuring high throughput for large-scale data cleaning tasks.

D BATCH DELETION IN HYBRID

Algorithm 5: Batch Deletion

Input : Current dataset D_r, loss function $\mathcal{L}$, batch size k

Output: A removable batch $C^\star$

$C \leftarrow$ the k tuples with the smallest values of $\mathcal{L}(D_r \setminus \{t\})$, sorted in ascending order;

$\ell \leftarrow k$;

while $\ell > 1$ **do**

 if $\mathcal{L}(D_r \setminus C[1 : \ell]) < \mathcal{L}(D_r)$ **then**

  return $C[1 : \ell]$;

 $\ell \leftarrow \lceil \ell / 2 \rceil$;

return $C[1]$;

Algorithm 5 replaces the single-tuple deletion performed in Lines 6–8 of Algorithm 1. It first ranks tuples according to their individual removal loss, $\mathcal{L}(D_r \setminus \{t\})$, and retains the k best candidates. Since the candidates are ordered by increasing loss, the algorithm repeatedly tests whether removing the current prefix decreases the objective. If so, the entire prefix is removed; otherwise, the prefix size is halved and tested again.

E ALGORITHM 6 ANALYSIS

E.1 Pseudocode

Algorithm 6 presents the pseudocode of the private marginal selection and release procedure described in Section 5.

E.2 Privacy and Utility Analysis

We next analyze Algorithm 6, showing that it satisfies zCDP and its error bound. We then prove the sensitivity of the utility function in Definition 20.

E.3 Proof to Proposition 19

Proof. **(1) Differential Privacy.** Algorithm 6 proceeds in two sequential stages: marginal selection (Lines 1–7) and noisy measurement computation (Line 11).

The first stage selects k marginals using the Exponential Mechanism [43]. Following established results on the adaptive composition of the Exponential Mechanism [6, 12, 13], this sequence satisfies ρ_{Topk}-zCDP. We stress that this stage relies on the *bounded-range* analysis of the EM, not on the generic $\frac{1}{2}\epsilon^2$-zCDP bound recorded in Section 2.2. An EM invocation with parameter ϵ is ϵ-bounded-range and therefore satisfies the strictly tighter $\frac{1}{8}\epsilon^2$-zCDP [6, 12, 51]; it is

Algorithm 6: Obtain k representative noisy marginals

input : Private and synthetic datasets D_p, D_s, a utility function u, a privacy parameter for top-k ρ_{Topk}, a number of desired marginals k, and a privacy parameter for Gaussian marginal measurement ρ_{marg}.

output : The monitored set $\mathcal{M}$: for each of the k selected attribute pairs (A_i, A_j), all of its value-pair entries with their noisy private estimates $\tilde{m}_{D_p}(A_i{=}a_i, A_j{=}a_j)$.

```
1  Compute all 2-way marginals of D_p and of D_s, m_D(i, j);
2  Let σ = 2Δ_u sqrt(k / (8ρ_Topk)), M = ∅ ;
3  foreach marginal m_{D_p}(i, j) do
4      score_u(m_{D_p}(i, j)) = u(m_{D_p}(i, j)) + Gumbel(σ);
5  end
6  Sort the marginals by score_u;
7  Let T_k be the top-k marginals by score_u;
   /* Gaussian Mechanism: add N(0, σ²_marg) noise to each
      marginal entry (ρ_marg/k-zCDP per marginal).        */
8  σ_marg ← √k / (n √ρ_marg);
9  foreach (A_i, A_j) ∈ T_k do
10     foreach (a_i, a_j) ∈ Dom(A_i) × Dom(A_j) do
11         m̃_{D_p}(A_i=a_i, A_j=a_j) ←
             m_{D_p}(A_i=a_i, A_j=a_j) + N(0, σ²_marg);
12     end
13     Clip each entry to [0, 1] and renormalize so entries sum
        to 1;
14     M ← M ∪ {(A_i=a_i, A_j=a_j, m̃_{D_p}(A_i=a_i, A_j=a_j))};
15 end
16 return M;
```

this $\frac{1}{8}$ constant, and not the $\frac{1}{2}$ constant of the preliminaries, that our accounting uses. Allocating ρ_{Topk}/k-zCDP to each of the k selections and inverting $\frac{1}{8}\epsilon^2 = \rho_{Topk}/k$ yields the per-step parameter $\epsilon = \sqrt{8\rho_{Topk}/k}$, which is exactly the value to which Algorithm 6 (Line 2) calibrates the Gumbel scale $\sigma = 2\Delta_u/\epsilon = 2\Delta_u\sqrt{k/(8\rho_{Topk})}$. Had we instead used the looser $\frac{1}{2}\epsilon^2$ bound of Section 2.2, the per-step parameter would be $\epsilon = \sqrt{2\rho_{Topk}/k}$ and the calibrated Gumbel scale would be larger by a factor of 2 (i.e., twice the injected noise), degrading the utility bound in Part (2) accordingly. The second stage applies the Gaussian Mechanism to each of the k selected marginals with per-marginal budget ρ_{marg}/k-zCDP (noise $\sigma_{\text{marg}} = \sqrt{k}/(n\sqrt{\rho_{marg}})$, calibrated to the ℓ_2 sensitivity $\Delta_2 = \sqrt{2}/n$ of the vector of all entries of a normalized marginal: replacing one tuple changes at most two entries by $1/n$ each; see Definition 20). By sequential composition of the k independent Gaussian measurements (Proposition 7), this stage satisfies ρ_{marg}-zCDP in total. By the sequential composition property of zCDP (Proposition 7), Algorithm 6 is $(\rho_{Topk} + \rho_{marg})$-zCDP.

(2) Utility Bound. We derive the utility bound by extending the standard utility theorem for the Exponential Mechanism (EM) [43], as presented in [17]. Denote the exponential mechanism by EM. Recall that for a single execution with privacy budget ε, the probability that the EM outputs a result $r = EM(D, \mathcal{R})$ (with candidates set $\mathcal{R}$) with utility $u(D, r)$ significantly below the optimal utility OPT is bounded by:

$$\Pr\left[u(r) \le OPT - \frac{2\Delta_u}{\varepsilon}(\ln(|\mathcal{R}|) + t)\right] \le e^{-t} \tag{8}$$

where Δ_u is the sensitivity of the utility function. To adapt this to our setting of k selections over the candidates set of all the marginals $\mathcal{M}_0$, we define the effective per-step budget $\varepsilon = \sqrt{8\rho_{Topk}/k}$ (the bounded-range parameter derived in Part (1) by inverting $\frac{1}{8}\varepsilon^2 = \rho_{Topk}/k$; note this is the $\frac{1}{8}\varepsilon^2$-zCDP bound of the EM [6, 12, 51], not the $\frac{1}{2}\varepsilon^2$ bound of Section 2.2).

Before applying this bound per step, we justify treating the one-shot selection of Algorithm 6 as k sequential EM steps. The algorithm perturbs every candidate's utility with a single draw of Gumbel noise and returns the k highest-scoring marginals (Lines 4–7). Because the Exponential Mechanism is equivalent to Report-Noisy-Max with Gumbel noise, this one-shot top-k has the same joint distribution as the sequential "peeling" process that runs k Exponential Mechanisms, where the i-th mechanism selects from the candidates not yet chosen [13, 14]. Consequently, the marginal returned at index i is distributed exactly as the i-th sequential EM over the remaining candidates, so the single-execution tail bound above applies verbatim to each index.

Let $OPT^{(i)}$ denote the optimal utility available at the i-th step, and $\mathcal{M}_0^{(i)}$ the marginal selected at that step. Following the proof structure in [17], for any utility threshold s, the probability of selecting a candidate with utility at most s is bounded by:

$$\Pr\left[u(\mathcal{M}_0^{(i)}) \le s\right] \le \frac{|\mathcal{M}_0| \exp\left(\frac{\varepsilon s}{2\Delta_u}\right)}{\exp\left(\frac{\varepsilon OPT^{(i)}}{2\Delta_u}\right)} = |\mathcal{M}_0| \exp\left(\frac{\varepsilon}{2\Delta_u}(s - OPT^{(i)})\right) \tag{9}$$

Substituting $s = OPT^{(i)} - \frac{2\Delta_u}{\varepsilon}(\ln(|\mathcal{M}_0|) + t)$ into the inequality yields the desired bound for each step $i \in \{1, \ldots, k\}$. □

E.4 Proof to Proposition 21

Proof. Fix an attribute pair (A_i, A_j) and write $c_{D_p}(a_i, a_j) = |\{t \in D_p : t[A_i] = a_i \wedge t[A_j] = a_j\}|$ for the cell counts, so that $m_{D_p}(A_i{=}a_i, A_j{=}a_j) = c_{D_p}(a_i, a_j)/n$ with $n = |D_p|$. By Definition 20,

$$u(A_i, A_j) = \sum_{(a_i, a_j)} \left|\frac{c_{D_p}(a_i, a_j)}{n} - m_{D_s}(A_i{=}a_i, A_j{=}a_j)\right|.$$

Let D_p and D'_p be neighboring private datasets (Definition 5), i.e., D'_p is obtained from D_p by replacing one tuple t with a tuple t'; the synthetic marginals $m_{D_s}(\cdot,\cdot)$ and the size n are unchanged. Applying the triangle inequality $\big||x| - |y|\big| \le |x - y|$ to each cell and summing,

$$|u_{D_p}(A_i, A_j) - u_{D'_p}(A_i, A_j)| \le \frac{1}{n} \sum_{(a_i, a_j)} \left|c_{D_p}(a_i, a_j) - c_{D'_p}(a_i, a_j)\right|.$$

Replacing t with t' decreases the count of the cell $(t[A_i], t[A_j])$ by one and increases the count of the cell $(t'[A_i], t'[A_j])$ by one, changing no other cell, so the sum on the right-hand side is at most 2. Hence $\Delta_u \le 2/n$, and the bound is attained when t and t' fall in different cells and both cell differences push u in the same direction. Therefore $\Delta_u = \frac{2}{n}$. □

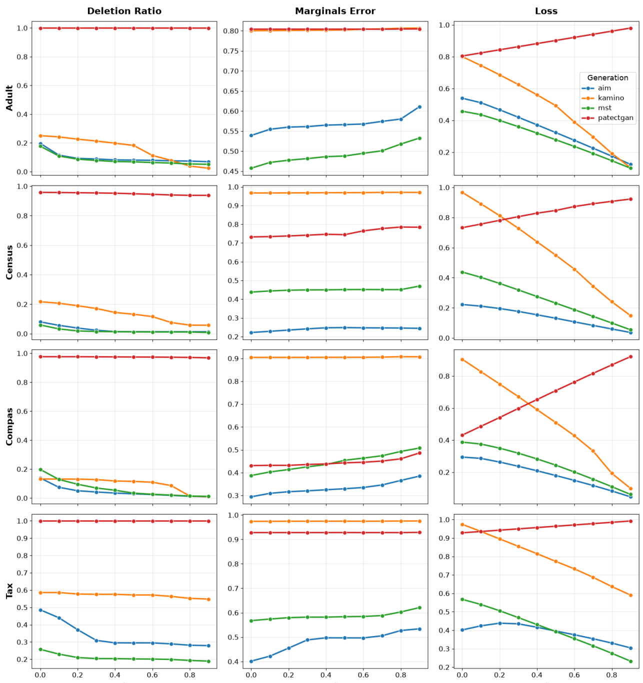


Figure 9: Deletion ratio (left) marginal error (center) and loss (right) of Hybrid as function of the tradeoff parameter α.

Table 6: Full list of denial constraints per dataset used in Section 6.

Dataset	Constraints
IPUMS-CPS	(1) $\forall t_1, t_2 \in D : \neg(t_1.AGEP = 0 \wedge t_2.AGEP = 0 \wedge t_1.SCHL \neq t_2.SCHL)$ (2) $\forall t_1, t_2 \in D : \neg(t_1.AGEP = 0 \wedge t_2.AGEP = 0 \wedge t_1.WAGP \neq t_2.WAGP)$ (3) $\forall t_1, t_2 \in D : \neg(t_1.CIT = t_2.CIT \wedge t_1.NATIVITY \neq t_2.NATIVITY)$ (4) $\forall t_1, t_2 \in D : \neg(t_1.AGEP = 0 \wedge t_2.AGEP = 0 \wedge t_1.MAR \neq t_2.MAR)$ (5) $\forall t_1, t_2 \in D : \neg(t_1.ST = t_2.ST \wedge t_1.DIVISION \neq t_2.DIVISION)$
Adult	(1) $\forall t_1, t_2 \in D : \neg(t_1.education\_num \neq t_2.education\_num \wedge t_1.education = t_2.education)$ (2) $\forall t_1, t_2 \in D : \neg(t_1.education\_num = t_2.education\_num \wedge t_1.education \neq t_2.education)$ (3) $\forall t_1, t_2 \in D : \neg(t_1.capital\_gain < t_2.capital\_gain \wedge t_1.capital\_loss < t_2.capital\_loss)$
Compas	(1) $\forall t_1, t_2 \in D : \neg(t_1.DecileScore = t_2.DecileScore \wedge t_1.ScoreText \neq t_2.ScoreText)$ (2) $\forall t_1, t_2 \in D : \neg(t_1.Scale\_ID = t_2.Scale\_ID \wedge t_1.DisplayText \neq t_2.DisplayText)$ (3) $\forall t_1, t_2 \in D : \neg(t_1.RecSupervisionLevel = t_2.RecSupervisionLevel \wedge t_1.RecSupervisionLevelText \neq t_2.RecSupervisionLevelText)$ (4) $\forall t_1, t_2 \in D : \neg(t_1.DisplayText = t_2.DisplayText \wedge t_1.Scale\_ID \neq t_2.Scale\_ID)$ (5) $\forall t_1, t_2 \in D : \neg(t_1.RecSupervisionLevelText = t_2.RecSupervisionLevelText \wedge t_1.RecSupervisionLevel \neq t_2.RecSupervisionLevel)$
Tax	(1) $\forall t_1, t_2 \in D : \neg(t_1.State = t_2.State \wedge t_1.HasChild = t_2.HasChild \wedge t_1.ChildExemp \neq t_2.ChildExemp)$ (2) $\forall t_1, t_2 \in D : \neg(t_1.State = t_2.State \wedge t_1.MaritalStatus = t_2.MaritalStatus \wedge t_1.SingleExemp \neq t_2.SingleExemp)$ (3) $\forall t_1, t_2 \in D : \neg(t_1.State = t_2.State \wedge t_1.Salary > t_2.Salary \wedge t_1.Rate < t_2.Rate)$ (4) $\forall t_1, t_2 \in D : \neg(t_1.State = t_2.State \wedge t_1.Salary < t_2.Salary \wedge t_1.Rate > t_2.Rate)$

F DENIAL CONSTRAINTS USED IN SECTION 6

The full set of DCs per dataset appears in Table 6. Most of the DCs are taken from previous work [36, 47, 50] and express innate semantic dependencies between attributes in the datasets. Some new constraints were discovered due to bucketization. Specifically, for [19], we bucketize AGEP and WAGP which represent the age and income of a person. Thus, for example, the first DC in Table 6 expresses that two tuples in a specific age group (coded by the bucket 0) have the same years of schooling. This constraint was examined and found to be satisfied by the data.

G α PLOTS

Figure 9 reports the deletion ratio (left), mean marginal error (center), and optimization loss (right) of Hybridas a function of the tradeoff parameter α. As expected, increasing α consistently decreases the deletion ratio while slightly increasing the marginal error. This follows directly from the loss function introduced in Definition 11,

$$\mathcal{L}(D_r) = \alpha \frac{n - |D_r|}{n} + (1 - \alpha) \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} E_m(m, D_r),$$

where $E_m(m, D_r)$ denotes the relative error of marginal m on the repaired dataset D_r. Larger values of α place more emphasis on preserving tuples and less emphasis on improving the marginal statistics.

This behavior can also be seen from the acceptance criterion of the repair algorithm. Since every accepted deletion must decrease the loss, if removing k tuples transforms $D_r^{(i)}$ into $D_r^{(i+1)}$, then

$$\mathcal{L}(D_r^{(i+1)}) < \mathcal{L}(D_r^{(i)}).$$

Rearranging the loss yields

$$\frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \left(E_m(m, D_r^{(i)}) - E_m(m, D_r^{(i+1)}) \right) > \frac{\alpha}{1 - \alpha} \cdot \frac{k}{n}$$

Therefore, every deletion must improve the average marginal error by at least $\frac{\alpha}{1-\alpha} \cdot \frac{k}{n}$. Since this threshold is monotonically increasing in α, larger values of α make deletions progressively harder to justify, resulting in fewer removed tuples and slightly larger marginal error.

The extreme case $\alpha = 1$ corresponds exactly to the classical repair objective,

$$\mathcal{L}(D_r) = \frac{n - |D_r|}{n},$$

which ignores marginal quality altogether. Consequently, after the first phase establishes a feasible repair, no further deletion can reduce the objective, and the marginal-improvement phase terminates immediately.

Empirically, however, the effect of α is relatively modest across all datasets. Although the threshold $\frac{\alpha}{1-\alpha} \frac{k}{n}$ grows rapidly as α approaches one, in practice $k \ll n$, so the required improvement in marginal error remains small over the range of α considered in our experiments. As a result, the deletion ratio and marginal error vary only gradually.

The optimization loss changes approximately linearly with α, which is expected because both the deletion ratio and the marginal error remain nearly constant throughout the sweep. In this regime, the loss is approximately an affine function of α with slope

$$\frac{n - |D_r|}{n} - \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} E_m(m, D_r).$$

For PATE-CTGAN, the deletion ratio is substantially larger due to the large number of constraint violations, yielding a positive slope and an increasing loss. In contrast, AIM, MST, and Kamino exhibit much lower deletion ratios, resulting in a negative slope and a decreasing loss as α increases.

H ρ PLOTS

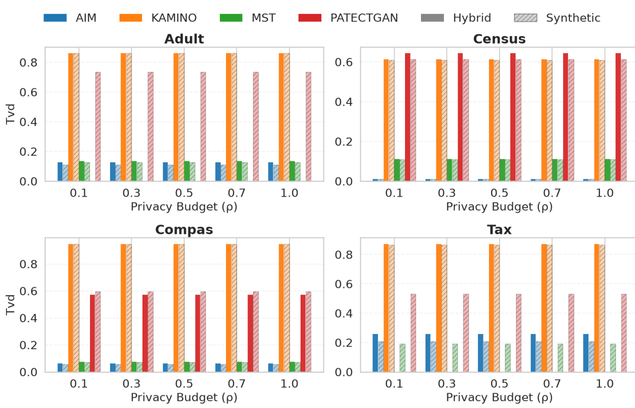


Figure 10: TVD of repaired data using HYBRID as function of privacy budget given to the marginals.

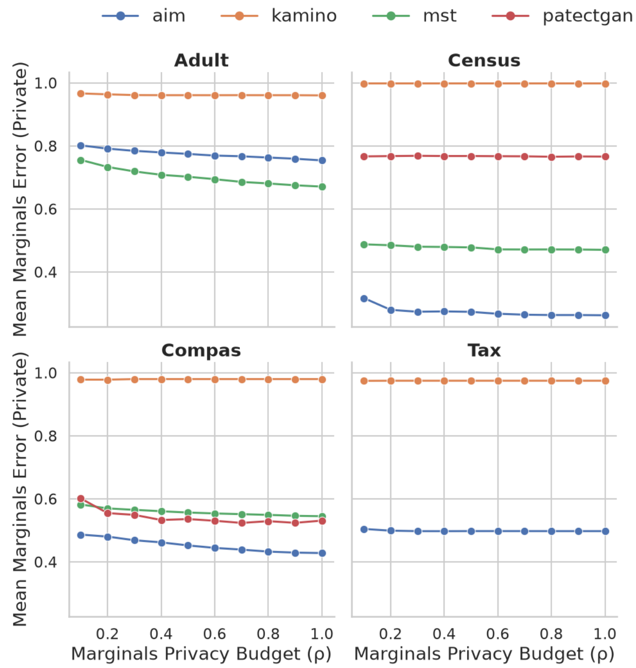


Figure 11: Marginals error with respect to the private data of repaired data using HYBRID as function of privacy budget given to the marginals.

Figure 10 and Figure 11 report the two-way TVD and mean marginal error with respect to the private data as a function of the privacy budget allocated to marginal estimation. As described in Section 6.1, the target marginals are selected uniformly at random and estimated using the Gaussian mechanism. These experiments evaluate how the accuracy of the noisy marginals affects the quality of the repaired dataset.

As expected, repair slightly increases the two-way TVD compared to the original synthetic data. This is because the optimization explicitly targets the selected marginals rather than the full joint distribution, so improving the selected marginals may slightly degrade global distributional similarity. Nevertheless, the increase is small and remains largely insensitive to the privacy budget. This is expected since only a subset of marginals is optimized, and improvements to their estimates have a limited effect on the overall TVD.

In contrast, the marginal error consistently decreases as the privacy budget increases. With a larger budget, the Gaussian mechanism adds less noise to the target marginals, providing more accurate optimization targets and allowing the repair algorithm to better match the corresponding statistics of the private data.